Lecture Notes: Statistical Learning

Martin Holler

Winter term 2026, Version: October 8, 2026

CONTENTS

I INTRODUCTION 5

II NOTATION 7

III STATISTICAL LEARNING 11

III.1 Supervised learning problems and loss functions 11

III.2 Expected risk 12

III.3 Empirical risk 13

III.4 Bayes predictor and Bayes risk 13

III.5 Excess risk 16

III.6 No free lunch 16

III.7 Hypothesis classes and the risk decomposition 17

III.8 Estimation error 19

III.8.1 Reduction to uniform deviations 19

III.8.2 A first concentration result: the law of large numbers 20

III.8.3 From bounds in expectation to bounds in high probability 21

III.8.4 Hoeffding's inequality 22

A RECAP: BASIC CONCEPTS FROM PROBABILITY THEORY 27

A.1 Random variables and their distribution 27

A.2 Expectation and moments 30

A.3 Distribution functions and quantiles 32

A.4 Independence, marginal and conditional distributions 34

A.5 Data: i.i.d. samples and exchangeability 41

I

INTRODUCTION

The main goal of statistical learning theory (for supervised learning) is as follows: Given a predictor $\hat{f} : \mathcal{X} \to \mathcal{Y}$ that was trained on data $(X_n, Y_n)_{n=1}^N$, drawn i.i.d. from an (unknown) joint distribution $\mathbb{P}_{X,Y}$ on $\mathcal{X} \times \mathcal{Y}$, what can we say about its performance on a new data point $(X_{\text{test}}, Y_{\text{test}})$ from the same distribution?

In this lecture, we address this question from two perspectives:

1. Classical statistical learning theory: How to estimate the generalization error of a given class of predictors?
2. Conformal prediction: How to obtain quantitative error bounds for black-box machine learning approaches using calibration data?

Note: This is not an introductory lecture to machine learning; the lecture assumes that you already have a basic knowledge of machine learning methods and concepts.

II

NOTATION

In this chapter, we briefly fix the notation used throughout the lecture. A more detailed recap of the underlying concepts from probability theory, including examples, can be found in Chapter A.

PROBABILITY SPACE AND DATA. All random quantities are defined on an underlying probability space $(\Omega, \Sigma, \mathbb{P})$ (see Section A.1), which we will hardly ever need explicitly. In supervised learning, a data point consists of an **input** X with values in an input space $\mathcal{X}$ and a **label** (or output) Y with values in a label space $\mathcal{Y}$. Typical examples are $\mathcal{X} = \mathbb{R}^d$, and $\mathcal{Y} = \{1, \dots, K\}$ for classification or $\mathcal{Y} = \mathbb{R}$ for regression. Formally, X and Y are random variables, i.e., (measurable) functions

$$X : \Omega \to \mathcal{X}, \qquad Y : \Omega \to \mathcal{Y},$$

and the data point is the random variable

$$(X, Y) : \Omega \to \mathcal{X} \times \mathcal{Y}, \qquad \omega \mapsto (X(\omega), Y(\omega)).$$

We use capital letters (X, Y) for random variables and lower-case letters (x, y) for fixed values (realizations) in $\mathcal{X}$ and $\mathcal{Y}$.

DISTRIBUTIONS. The **distribution** of X is the probability measure $\mathbb{P}_X$ on $\mathcal{X}$ given by

$$\mathbb{P}_X(A) = \mathbb{P}(X \in A) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \in A\}) \quad \text{for } A \subseteq \mathcal{X},$$

and we write $X \sim \mathbb{P}_X$. Similarly, the **joint distribution** $\mathbb{P}_{X,Y}$ of the pair (X, Y) is a probability measure on $\mathcal{X} \times \mathcal{Y}$ with

$$\mathbb{P}_{X,Y}(A \times B) = \mathbb{P}(X \in A, Y \in B) \quad \text{for } A \subseteq \mathcal{X},\ B \subseteq \mathcal{Y}.$$

The distributions $\mathbb{P}_X$ and $\mathbb{P}_Y$ of the input and the label alone are called the **marginal distributions** of $\mathbb{P}_{X,Y}$. In statistical learning, $\mathbb{P}_{X,Y}$ is **unknown**, and we only have access to samples from it.

DENSITIES. Whenever it exists, we describe a distribution by its **density** $p_X : \mathcal{X} \to [0, \infty)$, such that

$$\mathbb{P}(X \in A) = \sum_{x \in A} p_X(x) \quad (\mathcal{X} \text{ discrete}), \qquad \mathbb{P}(X \in A) = \int_A p_X(x)\, dx \quad (\mathcal{X} = \mathbb{R}^d).$$

Accordingly, $p_{X,Y}$ denotes the joint density of (X, Y) (also in the mixed case of continuous inputs and discrete labels, see Definition A.8), and

$$p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)} \qquad (\text{for } p_X(x) > 0)$$

the **conditional density** of Y given $X = x$. For classification, $p_{Y|X}(y|x)$ is the probability of class y given the input x.

EXPECTATIONS. We write $\mathbb{E}[g(X)]$ for the expectation of a function of X, $\mathbb{E}[g(X)|Y = y]$ for the conditional expectation given $Y = y$, and $\mathbb{1}\{\cdot\}$ for the indicator of a statement, such that $\mathbb{P}(X \in A) = \mathbb{E}[\mathbb{1}\{X \in A\}]$ (see Sections A.2 and A.4). For instance, the **risk** of a predictor $f : \mathcal{X} \to \mathcal{Y}$ with respect to a loss function ℓ is $\mathbb{E}[\ell(Y, f(X))]$ (see Chapter III).

DATA SETS. A data set $(X_n, Y_n)_{n=1}^N$ consists of N random variables that are typically assumed to be independent and identically distributed (i.i.d.) with distribution $\mathbb{P}_{X,Y}$, denoted $(X_n, Y_n) \overset{\text{i.i.d.}}{\sim} \mathbb{P}_{X,Y}$. A new data point, on which the performance of a predictor is evaluated, is denoted by $(X_{\text{test}}, Y_{\text{test}}) \sim \mathbb{P}_{X,Y}$ and assumed to be independent of the data set. In conformal prediction, the i.i.d. assumption will be relaxed to exchangeability (see Section A.5).

Example II.1 (Three dice)**.** *We throw three fair dice, i.e., $\Omega = \{1, \dots, 6\}^3$ with $\mathbb{P}(\{\omega\}) = \frac{1}{216}$ for each outcome $\omega = (d_1, d_2, d_3) \in \Omega$. As input, we observe the first two dice and only whether the third die is even or odd, and the label is the sum of all three dice:*

$$X(\omega) = (d_1, d_2, b) \quad \textit{with} \quad b = \begin{cases} 0 & \textit{if } d_3 \textit{ is even,} \\ 1 & \textit{if } d_3 \textit{ is odd,} \end{cases} \qquad Y(\omega) = d_1 + d_2 + d_3.$$

Hence $\mathcal{X} = \{1, \dots, 6\}^2 \times \{0, 1\}$ and $\mathcal{Y} = \{3, \dots, 18\}$.

- ***Density of** X: Each $(x_1, x_2) \in \{1, \dots, 6\}^2$ occurs with probability $\frac{1}{36}$, and the third die is even with probability $\frac{1}{2}$. Hence X is uniformly distributed on $\mathcal{X}$:*

$$p_X(x_1, x_2, b) = \frac{1}{36} \cdot \frac{1}{2} = \frac{1}{72} \quad \textit{for all } (x_1, x_2, b) \in \mathcal{X}, \quad \textit{e.g.,} \quad p_X(2, 2, 1) = \frac{1}{72}.$$

- ***Density of** Y: We count the outcomes ω with $d_1 + d_2 + d_3 = y$. For example, $y = 3$ is only obtained by $(1, 1, 1)$, $y = 5$ by the six permutations of $(1, 1, 3)$ and $(1, 2, 2)$, and $y = 10$ by 27 outcomes. Hence*

$$p_Y(3) = \frac{1}{216}, \qquad p_Y(5) = \frac{6}{216}, \qquad p_Y(10) = \frac{27}{216} = \frac{1}{8}.$$

  The density p_Y is symmetric around 10.5, with maximal value $\frac{1}{8}$ at $y \in \{10, 11\}$.

Our goal is to predict Y from X. Neither p_X nor p_Y tells us anything about how Y depends on X; this information is contained in the joint distribution $\mathbb{P}_{X,Y}$. Since at prediction time the input $X = x$ is

observed, everything that can be known about the label is contained in the ***conditional distribution*** $\mathbb{P}_{Y|X=x}$ *of* Y *given* $X = x$*, i.e., in the conditional density*

$$p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)}.$$

Knowing $p_{Y|X}(\cdot|x)$ *for all* $x \in \mathcal{X}$ *(together with* p_X*, since* $p_{X,Y}(x,y) = p_{Y|X}(y|x)p_X(x)$*) is thus the most complete information we can hope for: Any reasonable prediction, such as the most likely label, the expected label* $\mathbb{E}[Y|X = x]$*, or a set of plausible labels, can be derived from it.*

In the example, given $x = (x_1, x_2, b)$*, the label is* $Y = x_1 + x_2 + d_3$*, where* d_3 *is uniformly distributed on* $D_0 = \{2,4,6\}$ *if* $b = 0$ *and on* $D_1 = \{1,3,5\}$ *if* $b = 1$*. Hence*

$$p_{Y|X}(y|x_1, x_2, b) = \begin{cases} \frac{1}{3} & \text{if } y - x_1 - x_2 \in D_b, \\ 0 & \text{otherwise.} \end{cases}$$

Concretely, for $x = (2,2,1)$*, the third die is* 1*,* 3 *or* 5*. The joint density is* $p_{X,Y}((2,2,1), y) = \frac{1}{216}$ *for* $y \in \{5,7,9\}$ *and zero otherwise, such that*

$$p_{Y|X}(y|(2,2,1)) = \frac{1/216}{1/72} = \frac{1}{3} \quad \text{for } y \in \{5,7,9\}, \qquad \mathbb{E}[Y|X = (2,2,1)] = \frac{5+7+9}{3} = 7.$$

This shows that there is ***inherent uncertainty*** *in the data: Even with perfect knowledge of* $\mathbb{P}_{X,Y}$*, the label* Y *is not determined by the input* X*. For* $x = (2,2,1)$*, any prediction of a single label is correct with probability at most* $\frac{1}{3}$*, no matter how the predictor was obtained. A prediction set, in contrast, can quantify this uncertainty: the set* $\{5,7,9\}$ *contains the true label with probability one. Constructing such sets from data is the goal of conformal prediction.*

III

STATISTICAL LEARNING

III.1 SUPERVISED LEARNING PROBLEMS AND LOSS FUNCTIONS

As introduced in Chapter II, we consider a **feature space** (or input space) $\mathcal{X}$ and a **label space** (or output space) $\mathcal{Y}$, as well as an unknown joint distribution $\mathbb{P}_{X,Y}$ of features $X \in \mathcal{X}$ and labels $Y \in \mathcal{Y}$. The goal of supervised learning is to find a **predictor** (or prediction function)

$$f : \mathcal{X} \to \mathcal{Y},$$

such that $f(X)$ is a good prediction of Y. To quantify what "good" means, we need a **loss function**

$$\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R} \quad \text{(typically with values in } [0, \infty)),$$

where $\ell(y, z)$ is the loss incurred by predicting z when the true label is y. Note that the loss only concerns the label space and is independent of the feature space. Throughout the lecture, we assume that the loss function is given; in practice, it is chosen according to the application, as it determines how the performance of a predictor is measured.

Remark III.1. *Some authors swap the order of the arguments, i.e., write $\ell(z, y)$ for the loss of predicting z when the true label is y. We follow [1] and always use the true label as first argument.*

Example III.2 (Features, labels and losses)**.**

- ***Binary classification:*** *Features $\mathcal{X} = \mathbb{R}^d$, e.g., d medical measurements of a patient, and labels $\mathcal{Y} = \{0, 1\}$, e.g., whether the patient has a certain disease (1) or not (0). Often, also $\mathcal{Y} = \{-1, 1\}$ is used. The most common loss is the* ***0–1 loss***

$$\ell(y, z) = \mathbb{1}\{y \neq z\} = \begin{cases} 0 & \text{if } y = z \text{ (no mistake)}, \\ 1 & \text{if } y \neq z \text{ (mistake)}. \end{cases}$$

  If the two types of mistakes have different consequences, e.g., missing a disease is worse than a false alarm, an ***asymmetric loss*** *with $\ell(1, 0) = c > 1$, $\ell(0, 1) = 1$ and $\ell(0, 0) = \ell(1, 1) = 0$ can be used.*

- ***Multi-category classification:*** *Features $\mathcal{X} = \mathbb{R}^{h \times w \times 3}$, e.g., the RGB values of an image with $h \times w$ pixels, and labels $\mathcal{Y} = \{1, \dots, K\}$, e.g., the object shown in the image out of K categories. Again, the 0–1 loss $\ell(y, z) = \mathbb{1}\{y \neq z\}$ is most common.*

- ***Regression:*** *Features $\mathcal{X} = \mathbb{R}^d$, e.g., size, location and age of a house, and labels $\mathcal{Y} = \mathbb{R}$, e.g., its selling price. The most common loss is the* ***squared loss*** *$\ell(y, z) = (y - z)^2$. The* ***absolute loss***

$\ell(y,z) = |y-z|$ penalizes large errors less and is used if robustness with respect to outliers is important.

- ***Three dice:*** *In Example II.1, the features are $\mathcal{X} = \{1,\dots,6\}^2 \times \{0,1\}$ and the labels $\mathcal{Y} = \{3,\dots,18\}$. Since the labels are numbers, predicting the sum can be considered both as multi-category classification with the 0–1 loss (only exactly correct predictions count) and as regression with the squared loss (predictions close to the true sum are better than those far away).*

III.2 EXPECTED RISK

The loss measures the quality of a prediction for a single data point. Since data points are random, the quality of a predictor is measured by the average loss with respect to the distribution of the data.

Definition III.3 (Expected risk)**.** *Let $f : \mathcal{X} \to \mathcal{Y}$ be a predictor, $\ell : \mathcal{Y} \times \mathcal{Y} \to [0,\infty)$ a loss function, and (X,Y) a random variable with distribution $\mathbb{P}_{X,Y}$ on $\mathcal{X} \times \mathcal{Y}$. The **expected risk** (or simply **risk**) of f is defined as*

$$\mathcal{R}(f) = \mathbb{E}\big[\ell(Y, f(X))\big].$$

If (X,Y) has a joint density $p_{X,Y}$, then by Proposition A.14,

$$\mathcal{R}(f) = \int_{\mathcal{X}} \int_{\mathcal{Y}} \ell(y, f(x))\, p_{X,Y}(x,y)\, dy\, dx,$$

where integrals are replaced by sums for discrete variables.

Remark III.4.

- *The expected risk depends on the distribution $\mathbb{P}_{X,Y}$; if this needs to be made explicit, we write $\mathcal{R}_{\mathbb{P}_{X,Y}}(f)$. Since $\mathbb{P}_{X,Y}$ is unknown, the expected risk cannot be computed in practice; it is our main performance criterion that will be estimated and controlled using data (see Section III.3).*

- *For nonnegative losses, the risk is always well defined, but may be infinite.*

Example III.5 (Risk for classification and regression)**.**

- ***Binary classification:*** *For $\mathcal{Y} = \{0,1\}$ and the 0–1 loss, using $\mathbb{E}[\mathbb{1}\{\cdot\}] = \mathbb{P}(\cdot)$, the risk is*

  $$\mathcal{R}(f) = \mathbb{E}\big[\mathbb{1}\{Y \neq f(X)\}\big] = \mathbb{P}(f(X) \neq Y) = \mathbb{P}(f(X) = 1, Y = 0) + \mathbb{P}(f(X) = 0, Y = 1),$$

  *i.e., the probability of a mistake on a new data point, also called the **error rate**. It is the sum of the probabilities of the two types of errors, false positives and false negatives. In practice, often the **accuracy** $1 - \mathcal{R}(f) = \mathbb{P}(f(X) = Y)$ is reported.*

- ***Regression:*** *For $\mathcal{Y} = \mathbb{R}$, $\mathcal{X} = \mathbb{R}^d$ and the squared loss, the risk is the **mean squared error***

  $$\mathcal{R}(f) = \mathbb{E}\big[(Y - f(X))^2\big] = \int_{\mathbb{R}^d} \int_{\mathbb{R}} (y - f(x))^2\, p_{X,Y}(x,y)\, dy\, dx.$$

III.3 EMPIRICAL RISK

Since $\mathbb{P}_{X,Y}$ is unknown, the expected risk cannot be computed. What we have instead is data $(X_n, Y_n)_{n=1}^N$, assumed to be i.i.d. with distribution $\mathbb{P}_{X,Y}$. Replacing the expectation in the definition of the risk by an average over the data leads to the empirical risk.

Definition III.6 (Empirical risk)**.** *Let $f : \mathcal{X} \to \mathcal{Y}$ be a predictor, $\ell : \mathcal{Y} \times \mathcal{Y} \to [0, \infty)$ a loss function, and $(X_n, Y_n)_{n=1}^N$ data in $\mathcal{X} \times \mathcal{Y}$. The **empirical risk** (or **training error**) of f is defined as*

$$\hat{\mathcal{R}}_N(f) = \frac{1}{N} \sum_{n=1}^{N} \ell(Y_n, f(X_n)).$$

Remark III.7.

- *Since the data are random, so is $\hat{\mathcal{R}}_N(f)$. For a fixed predictor f, i.e., one that does not depend on the data, it is an unbiased estimate of the risk: By linearity of the expectation and since $(X_n, Y_n) \sim \mathbb{P}_{X,Y}$ for all n,*

$$\mathbb{E}\big[\hat{\mathcal{R}}_N(f)\big] = \frac{1}{N} \sum_{n=1}^{N} \mathbb{E}\big[\ell(Y_n, f(X_n))\big] = \mathcal{R}(f).$$

- *For classification with the 0–1 loss, $\hat{\mathcal{R}}_N(f) = \frac{1}{N}\#\{n : f(X_n) \neq Y_n\}$ is the proportion of mistakes on the data.*

The most basic learning approach is **empirical risk minimization**: Choose a class $\mathcal{F}$ of candidate predictors $f : \mathcal{X} \to \mathcal{Y}$ (e.g., linear functions or neural networks with a fixed architecture) and compute

$$\hat{f} \in \operatorname*{argmin}_{f \in \mathcal{F}} \hat{\mathcal{R}}_N(f).$$

Remark III.8 (The risk of a trained predictor is random)**.** *For a predictor $\hat{f}$ trained on data $(X_n, Y_n)_{n=1}^N$, the risk $\mathcal{R}(\hat{f})$ is the expected loss on a new data point $(X_{test}, Y_{test}) \sim \mathbb{P}_{X,Y}$ that is independent of the training data, i.e., it measures the performance on unseen data. It is therefore also called **generalization error** or **test error**. Since $\hat{f}$ depends on the training data, $\mathcal{R}(\hat{f})$ is itself a random variable (a function of the data), and so is $\hat{\mathcal{R}}_N(\hat{f})$. Moreover, the unbiasedness above no longer holds: In general, $\mathbb{E}[\hat{\mathcal{R}}_N(\hat{f})] \neq \mathbb{E}[\mathcal{R}(\hat{f})]$, as $\hat{f}$ was chosen to perform well on the same data that is used to evaluate $\hat{\mathcal{R}}_N(\hat{f})$. For example, if $\mathcal{F}$ is rich enough to contain a predictor that interpolates the data, i.e., $\hat{f}(X_n) = Y_n$ for all n, then $\hat{\mathcal{R}}_N(\hat{f}) = 0$, while $\mathcal{R}(\hat{f})$ can be large (overfitting).*

This leads to one of the main questions considered in this lecture:

> Given a predictor $\hat{f}$ that was trained by minimizing the empirical risk $\hat{\mathcal{R}}_N$, what can we say about its true (expected) risk $\mathcal{R}(\hat{f})$?

III.4 BAYES PREDICTOR AND BAYES RISK

Before studying what can be achieved with data, we ask a more basic question: **What is the best we can do in terms of the risk**, i.e., which predictor $f : \mathcal{X} \to \mathcal{Y}$ has the smallest risk if we knew the distribution $\mathbb{P}_{X,Y}$ (and were not restricted to a class $\mathcal{F}$)?

To answer this question, we use the law of total expectation (Section A.4), conditioning on the features X:

$$\mathcal{R}(f) = \mathbb{E}\big[\ell(Y, f(X))\big] = \mathbb{E}_X\Big[\mathbb{E}\big[\ell(Y, f(X)) \,\big|\, X\big]\Big] = \int_{\mathcal{X}} \mathbb{E}\big[\ell(Y, f(x)) \,\big|\, X = x\big]\, p_X(x)\, dx.$$

Here, we used that by definition of the conditional expectation, $\mathbb{E}[\ell(Y, f(X))|X = x] = \int_{\mathcal{Y}} \ell(y, f(x))\, p_{Y|X}(y|x)\, dy = \mathbb{E}[\ell(Y, f(x))|X = x]$, i.e., inside the conditional expectation, we can replace X by its known value x. This motivates to define the **conditional risk** of predicting $z \in \mathcal{Y}$ at the feature $x \in \mathcal{X}$ as

$$r(z|x) = \mathbb{E}\big[\ell(Y, z) \,\big|\, X = x\big] = \int_{\mathcal{Y}} \ell(y, z)\, p_{Y|X}(y|x)\, dy,$$

which is a deterministic function of z and x, such that

$$\mathcal{R}(f) = \int_{\mathcal{X}} r(f(x)|x)\, p_X(x)\, dx.$$

The key observation is that for each x, the integrand only depends on the single value $f(x)$. Hence, $\mathcal{R}(f)$ is minimized by minimizing the conditional risk $r(z|x)$ with respect to z **separately for each** x, i.e., by predicting at each x the value z that minimizes the expected loss given $X = x$.

Definition III.9 (Bayes predictor and Bayes risk)**.** *A **Bayes predictor** is a predictor $f^* : \mathcal{X} \to \mathcal{Y}$ satisfying*

$$f^*(x) \in \operatorname*{argmin}_{z \in \mathcal{Y}} \mathbb{E}\big[\ell(Y, z) \,\big|\, X = x\big] = \operatorname*{argmin}_{z \in \mathcal{Y}} r(z|x) \quad \text{for all } x \in \mathcal{X} \text{ with } p_X(x) > 0.$$

*The **Bayes risk** is defined as*

$$\mathcal{R}^* = \mathbb{E}_X\Big[\inf_{z \in \mathcal{Y}} \mathbb{E}\big[\ell(Y, z) \,\big|\, X\big]\Big] = \int_{\mathcal{X}} \inf_{z \in \mathcal{Y}} r(z|x)\, p_X(x)\, dx.$$

Proposition III.10. *For every predictor $f : \mathcal{X} \to \mathcal{Y}$ it holds that $\mathcal{R}(f) \geq \mathcal{R}^*$. If a Bayes predictor f^* exists, then $\mathcal{R}(f^*) = \mathcal{R}^*$, i.e., f^* minimizes the risk over all predictors.*

Proof. For every x, we have $r(f(x)|x) \geq \inf_{z \in \mathcal{Y}} r(z|x)$, and integrating with respect to p_X yields $\mathcal{R}(f) \geq \mathcal{R}^*$. For a Bayes predictor, $r(f^*(x)|x) = \min_{z \in \mathcal{Y}} r(z|x)$ for all x with $p_X(x) > 0$, and integrating yields $\mathcal{R}(f^*) = \mathcal{R}^*$. □

Remark III.11.

- *The Bayes predictor depends on the unknown distribution $\mathbb{P}_{X,Y}$ and can thus not be computed in practice. It serves as a benchmark for the performance of predictors, see Section III.5.*
- *In general, $\mathcal{R}^* > 0$, which reflects the inherent uncertainty in the data. For instance, in Example II.1 with the 0–1 loss, we have $r(z|x) = \mathbb{P}(Y \neq z|X = x) = 1 - p_{Y|X}(z|x) \geq \frac{2}{3}$ for all z and x, with equality for the three possible values of Y given x. Hence $\mathcal{R}^* = \frac{2}{3}$: Even the best predictor makes a mistake with probability $\frac{2}{3}$.*
- *Minimizing the risk over a restricted class $\mathcal{F}$ does in general not decouple into separate problems for each x, as the values $f(x)$ for different x are then coupled through the class $\mathcal{F}$.*

Example III.12 (Bayes predictor for binary classification). *Let $\mathcal{Y} = \{0,1\}$ and $\ell(y,z) = \mathbb{1}\{y \neq z\}$ be the 0–1 loss. We denote by*

$$\eta(x) = p_{Y|X}(1|x) = \mathbb{P}(Y = 1|X = x)$$

the probability of class 1 given the feature x, such that $p_{Y|X}(0|x) = 1 - \eta(x)$.

- ***Conditional risk:*** *For $z \in \{0,1\}$, summing over the two possible labels yields*

$$r(z|x) = \sum_{y\in\{0,1\}} \mathbb{1}\{y \neq z\}\, p_{Y|X}(y|x) = p_{Y|X}(1-z|x),$$

  i.e., $r(0|x) = \eta(x)$ and $r(1|x) = 1 - \eta(x)$.

- ***Bayes predictor:*** *Minimizing $r(z|x)$ over $z \in \{0,1\}$ means choosing the class with the larger conditional probability:*

$$f^*(x) = \begin{cases} 1 & \text{if } \eta(x) > \frac{1}{2}, \\ 0 & \text{if } \eta(x) < \frac{1}{2}, \end{cases}$$

  while for $\eta(x) = \frac{1}{2}$ both classes are optimal (so the Bayes predictor is not unique). In short, $f^*(x) \in \operatorname{argmax}_{y\in\{0,1\}} p_{Y|X}(y|x)$: predict the most likely class.

- ***Bayes risk:*** *Since $\min_z r(z|x) = \min\{\eta(x), 1-\eta(x)\}$, we obtain*

$$\mathcal{R}^* = \mathbb{E}\big[\min\{\eta(X), 1-\eta(X)\}\big],$$

  which is zero if and only if $\eta(X) \in \{0,1\}$ with probability one, i.e., if the label is (almost surely) a deterministic function of the feature.

Example III.13 (Bayes predictor for regression). *Let $\mathcal{Y} = \mathbb{R}$ and $\ell(y,z) = (y-z)^2$ be the squared loss, and assume $\mathbb{E}[Y^2] < \infty$. We denote by $m(x) = \mathbb{E}[Y|X = x]$ the regression function (see Section A.4).*

- ***Conditional risk:*** *For $z \in \mathbb{R}$, we write $Y - z = (Y - m(x)) + (m(x) - z)$ and expand the square:*

$$(Y-z)^2 = (Y-m(x))^2 + 2(Y-m(x))(m(x)-z) + (m(x)-z)^2.$$

  Taking the conditional expectation given $X = x$ and using its linearity (Proposition A.44), where $m(x)$ and z are constants, yields

$$r(z|x) = \mathbb{E}\big[(Y-m(x))^2\big|X=x\big] + 2(m(x)-z)\,\mathbb{E}\big[Y-m(x)\big|X=x\big] + (m(x)-z)^2.$$

  *The middle term vanishes, since $\mathbb{E}[Y - m(x)|X = x] = \mathbb{E}[Y|X = x] - m(x) = 0$. With the **conditional variance** $\operatorname{var}(Y|X = x) = \mathbb{E}[(Y - m(x))^2|X = x]$, we obtain*

$$r(z|x) = \operatorname{var}(Y|X=x) + (m(x)-z)^2.$$

- ***Bayes predictor:*** *The first term does not depend on z, and the second term is minimal (namely zero) if and only if $z = m(x)$. Hence the Bayes predictor is unique (for $p_X(x) > 0$) and equal to the regression function:*

$$f^*(x) = m(x) = \mathbb{E}[Y|X=x].$$

- ***Bayes risk:*** *Inserting $z = m(x)$ yields $\min_z r(z|x) = \operatorname{var}(Y|X = x)$, such that the Bayes risk is the expected conditional variance:*

$$\mathcal{R}^* = \mathbb{E}\big[\operatorname{var}(Y|X)\big] = \int_{\mathcal{X}} \operatorname{var}(Y|X = x)\, p_X(x)\, dx.$$

Remark III.14. *The Bayes predictors above are specific to the 0–1 loss and the squared loss. Other losses lead to other Bayes predictors: For instance, the absolute loss $\ell(y, z) = |y - z|$ leads to the conditional median, and an asymmetric loss in binary classification leads to a threshold different from $\frac{1}{2}$ for $\eta(x)$.*

III.5 EXCESS RISK

The risk $\mathcal{R}(f)$ alone does not tell us whether a predictor is good or bad because it mixes two effects: the inherent uncertainty in the data, which no predictor can overcome and which is measured by $\mathcal{R}^*$, and the suboptimality of the predictor f. To assess the quality of f (and of the method that produced it), we need to separate the two by comparing to the best possible risk.

Definition III.15 (Excess risk)**.** *The **excess risk** of a predictor $f : \mathcal{X} \to \mathcal{Y}$ is*

$$\mathcal{R}(f) - \mathcal{R}^*.$$

By Proposition III.10, it is always nonnegative, and it is zero for Bayes predictors.

To discuss what can be achieved by learning from data, we formalize the notion of a learning method.

Definition III.16 (Learning algorithm)**.** *A **learning algorithm** $\mathcal{A}$ is a map that assigns to each data set $\mathcal{D}_N = ((x_1, y_1), \dots, (x_N, y_N)) \in (\mathcal{X} \times \mathcal{Y})^N$, $N \in \mathbb{N}$, a predictor $\mathcal{A}(\mathcal{D}_N) : \mathcal{X} \to \mathcal{Y}$.*

For instance, empirical risk minimization over a class $\mathcal{F}$ defines a learning algorithm via $\mathcal{A}(\mathcal{D}_N) \in \operatorname{argmin}_{f \in \mathcal{F}} \hat{\mathcal{R}}_N(f)$. If $\mathcal{D}_N = ((X_n, Y_n))_{n=1}^N$ consists of i.i.d. samples from a distribution P on $\mathcal{X} \times \mathcal{Y}$, then $\hat{f} = \mathcal{A}(\mathcal{D}_N)$ is random, and so is its excess risk $\mathcal{R}_P(\hat{f}) - \mathcal{R}_P^*$, where we now write $\mathcal{R}_P$ and $\mathcal{R}_P^*$ to make the dependence on the distribution explicit. A natural measure of the quality of $\mathcal{A}$ for the distribution P is the **expected excess risk**

$$\mathbb{E}\big[\mathcal{R}_P(\mathcal{A}(\mathcal{D}_N))\big] - \mathcal{R}_P^*,$$

where the expectation is taken with respect to the data $\mathcal{D}_N$. The goal of learning theory is to show that this quantity is small (and to understand how it decreases as N grows).

III.6 NO FREE LUNCH

Since the Bayes predictor is explicitly known once $\mathbb{P}_{X,Y}$ is known, it is tempting to look for a **universal** learning algorithm, which achieves a small expected excess risk for every distribution P, provided that N is sufficiently large (for a given, fixed N). The following result shows that this is impossible: For any algorithm and any number of data points, there is a distribution for which the algorithm is not better than random guessing. Learning is only possible with assumptions on the distribution.

Proposition III.17 (No free lunch – fixed N, [1, Proposition 2.2])**.** *Consider binary classification, i.e., $\mathcal{Y} = \{0,1\}$, with the 0–1 loss, and assume that $\mathcal{X}$ is infinite. Let $\mathcal{P}$ denote the set of all probability distributions on $\mathcal{X} \times \{0,1\}$. Then, for any $N \in \mathbb{N}$ and any learning algorithm $\mathcal{A}$,*

$$\sup_{P \in \mathcal{P}} \left(\mathbb{E}\left[\mathcal{R}_P(\mathcal{A}(\mathcal{D}_N))\right] - \mathcal{R}_P^* \right) \geq \frac{1}{2},$$

where, for each P, $\mathcal{D}_N$ consists of N i.i.d. samples from P.

Interpretation: A predictor that ignores the feature and guesses the label by a fair coin flip has risk $\frac{1}{2}$ for every distribution. The result states that for any algorithm and any N, there are distributions with $\mathcal{R}_P^* = 0$ (the label is a deterministic function of the feature) for which the expected risk of the algorithm is arbitrarily close to this chance level.

Consequence: Meaningful guarantees on the risk of a learned predictor require **assumptions** on the data distribution, e.g., that the Bayes predictor is well approximated by a class $\mathcal{F}$ of moderate size, or that η or m are smooth. Such assumptions will be a central ingredient of the results in the following chapters.

III.7 HYPOTHESIS CLASSES AND THE RISK DECOMPOSITION

The no free lunch theorem (Proposition III.17) shows that we need to make some assumption in order to obtain guarantees for learning. Usually, this assumption is that we restrict ourselves to a class $\mathcal{F}$ of predictors $f : \mathcal{X} \to \mathcal{Y}$, called **hypothesis class**, and that predictors in $\mathcal{F}$ are able to perform well, i.e., to approximate the Bayes predictor sufficiently well.

Restricting to a class $\mathcal{F}$ is also necessary for practical purposes: We want to implement the predictor on the computer, which requires that it can be described by finitely many numbers. Hence, $\mathcal{F}$ is typically a parametrized class $\mathcal{F} = \{f_\theta : \theta \in \Theta\}$ with a parameter set $\Theta \subseteq \mathbb{R}^p$.

Example III.18 (Hypothesis classes)**.**

- ***Linear regression:*** *For $\mathcal{X} = \mathbb{R}^d$ and $\mathcal{Y} = \mathbb{R}$, the class of linear predictors with parameter vector $\theta \in \mathbb{R}^d$ is*
$$\mathcal{F} = \{f_\theta : \theta \in \mathbb{R}^d\}, \qquad f_\theta(x) = \theta^\top x.$$

- ***One-hidden-layer feed-forward neural networks:*** *For $\mathcal{X} = \mathbb{R}^d$ and $\mathcal{Y} = \mathbb{R}$, an activation function $\sigma : \mathbb{R} \to \mathbb{R}$ (e.g., the ReLU $\sigma(t) = \max\{t, 0\}$) and a number $m \in \mathbb{N}$ of hidden neurons, we consider*
$$\mathcal{F} = \{\mathcal{N}_\theta : \theta \in \mathbb{R}^{m(d+2)}\}, \qquad \mathcal{N}_\theta(x) = \sum_{j=1}^{m} a_j \, \sigma(w_j^\top x + b_j),$$
with parameters $\theta = (a_j, w_j, b_j)_{j=1}^m$, where $a_j, b_j \in \mathbb{R}$ and $w_j \in \mathbb{R}^d$.

A second limitation is that we can only learn the predictor from finitely many data points $(X_n, Y_n)_{n=1}^N$. Hence, we cannot compute the best predictor in $\mathcal{F}$ (which would require knowledge of $\mathbb{P}_{X,Y}$), but only a predictor $\hat{f} \in \mathcal{F}$ obtained from the data, e.g., by minimizing the

empirical risk $\hat{\mathcal{R}}_N$ over $\mathcal{F}$. Both limitations are reflected in the following decomposition of the excess risk of $\hat{f}$:

$$\mathcal{R}(\hat{f}) - \mathcal{R}^* = \underbrace{\Big(\mathcal{R}(\hat{f}) - \inf_{f\in\mathcal{F}} \mathcal{R}(f)\Big)}_{\text{estimation error}} + \underbrace{\Big(\inf_{f\in\mathcal{F}} \mathcal{R}(f) - \mathcal{R}^*\Big)}_{\text{approximation error}}.$$

Both terms are nonnegative, and they have the following meaning:

- The **approximation error** measures how well the Bayes predictor can be approximated by predictors in $\mathcal{F}$. It is due to the restriction to $\mathcal{F}$, does not depend on the data (it is deterministic), and decreases if $\mathcal{F}$ is enlarged. If $\mathcal{F}$ is too small, the approximation error is large, and even the best predictor in $\mathcal{F}$ performs poorly (**underfitting**).

- The **estimation error** measures how much worse the learned predictor $\hat{f}$ is compared to the best predictor in $\mathcal{F}$. It is due to the fact that we only have finitely many data points, and it is random, since $\hat{f}$ depends on the data. It typically decreases as the number of data points N grows, and increases if $\mathcal{F}$ is enlarged: A rich class $\mathcal{F}$ contains predictors that fit the data, including its random fluctuations, very well, such that minimizing the empirical risk may result in a predictor with small empirical risk but large risk (**overfitting**, see also Section III.3).

In statistical learning theory, one usually considers classes $\mathcal{F} = \mathcal{F}_m$ that depend on a parameter m measuring their complexity, e.g., the number m of hidden neurons in the neural network example above. One then aims to bound the two terms separately:

- The approximation error is bounded as

$$\inf_{f\in\mathcal{F}_m} \mathcal{R}(f) - \mathcal{R}^* \le E_{\text{approx}}(m), \qquad \text{with} \quad E_{\text{approx}}(m) \to 0 \quad \text{as } m \to \infty.$$

- The estimation error is bounded (in expectation or with high probability) as

$$\mathcal{R}(\hat{f}) - \inf_{f\in\mathcal{F}_m} \mathcal{R}(f) \le E_{\text{est}}(m, N), \qquad \text{with} \quad E_{\text{est}}(m, N) \to 0 \quad \text{as } N \to \infty$$

  for fixed m, where $E_{\text{est}}(m, N)$ increases with the complexity m.

Together, this yields a bound on the excess risk:

$$\mathcal{R}(\hat{f}) - \mathcal{R}^* \le E_{\text{approx}}(m) + E_{\text{est}}(m, N).$$

The goal is to choose the complexity m such that these two terms are balanced: If m is too small, the approximation error dominates (underfitting); if m is too large for the given number of data points N, the estimation error dominates (overfitting). Choosing $m = m(N)$ appropriately, i.e., increasing with N but slowly enough that $E_{\text{est}}(m(N), N) \to 0$, results in a small overall excess risk that tends to zero as $N \to \infty$.

III.8 ESTIMATION ERROR

Following the risk decomposition of Section III.7, we analyze the two error terms separately. We start by considering the estimation error

$$\mathcal{R}(\hat{f}) - \inf_{f \in \mathcal{F}} \mathcal{R}(f),$$

where $\hat{f}$ is a predictor learned from the data $(X_n, Y_n)_{n=1}^N$.

III.8.1 *Reduction to uniform deviations*

Recall that for a predictor $f : \mathcal{X} \to \mathcal{Y}$ and a loss function ℓ, the (expected) risk and the empirical risk are given as

$$\mathcal{R}(f) = \mathbb{E}\big[\ell(Y, f(X))\big], \qquad \hat{\mathcal{R}}_N(f) = \frac{1}{N} \sum_{n=1}^{N} \ell(Y_n, f(X_n)),$$

where $(X, Y) \sim \mathbb{P}_{X,Y}$ and $(X_n, Y_n)_{n=1}^N$ are i.i.d. with distribution $\mathbb{P}_{X,Y}$. Throughout this section, let $\hat{f} \in \mathcal{F}$ be an empirical risk minimizer, i.e., $\hat{\mathcal{R}}_N(\hat{f}) \leq \hat{\mathcal{R}}_N(g)$ for all $g \in \mathcal{F}$.

Proposition III.19. *The estimation error of the empirical risk minimizer $\hat{f}$ satisfies*

$$\mathcal{R}(\hat{f}) - \inf_{f \in \mathcal{F}} \mathcal{R}(f) \leq \sup_{f \in \mathcal{F}} \big(\mathcal{R}(f) - \hat{\mathcal{R}}_N(f)\big) + \sup_{f \in \mathcal{F}} \big(\hat{\mathcal{R}}_N(f) - \mathcal{R}(f)\big) \leq 2 \sup_{f \in \mathcal{F}} \big|\mathcal{R}(f) - \hat{\mathcal{R}}_N(f)\big|.$$

Proof. Let $g \in \mathcal{F}$ be arbitrary. Adding and subtracting $\hat{\mathcal{R}}_N(\hat{f})$ and $\hat{\mathcal{R}}_N(g)$, we can write

$$\mathcal{R}(\hat{f}) - \mathcal{R}(g) = \big(\mathcal{R}(\hat{f}) - \hat{\mathcal{R}}_N(\hat{f})\big) + \big(\hat{\mathcal{R}}_N(\hat{f}) - \hat{\mathcal{R}}_N(g)\big) + \big(\hat{\mathcal{R}}_N(g) - \mathcal{R}(g)\big).$$

We estimate the three terms separately:

- Since $\hat{f} \in \mathcal{F}$, the first term satisfies $\mathcal{R}(\hat{f}) - \hat{\mathcal{R}}_N(\hat{f}) \leq \sup_{f \in \mathcal{F}} \big(\mathcal{R}(f) - \hat{\mathcal{R}}_N(f)\big)$.
- Since $\hat{f}$ minimizes $\hat{\mathcal{R}}_N$ over $\mathcal{F}$ and $g \in \mathcal{F}$, the second term satisfies $\hat{\mathcal{R}}_N(\hat{f}) - \hat{\mathcal{R}}_N(g) \leq 0$.
- Since $g \in \mathcal{F}$, the third term satisfies $\hat{\mathcal{R}}_N(g) - \mathcal{R}(g) \leq \sup_{f \in \mathcal{F}} \big(\hat{\mathcal{R}}_N(f) - \mathcal{R}(f)\big)$.

Hence

$$\mathcal{R}(\hat{f}) - \mathcal{R}(g) \leq \sup_{f \in \mathcal{F}} \big(\mathcal{R}(f) - \hat{\mathcal{R}}_N(f)\big) + \sup_{f \in \mathcal{F}} \big(\hat{\mathcal{R}}_N(f) - \mathcal{R}(f)\big).$$

Since the right-hand side does not depend on g, we can take the supremum over $g \in \mathcal{F}$ on the left-hand side, and using $\sup_{g \in \mathcal{F}} \big(\mathcal{R}(\hat{f}) - \mathcal{R}(g)\big) = \mathcal{R}(\hat{f}) - \inf_{g \in \mathcal{F}} \mathcal{R}(g)$, we obtain the first inequality. The second inequality follows since $\mathcal{R}(f) - \hat{\mathcal{R}}_N(f) \leq |\mathcal{R}(f) - \hat{\mathcal{R}}_N(f)|$ and $\hat{\mathcal{R}}_N(f) - \mathcal{R}(f) \leq |\mathcal{R}(f) - \hat{\mathcal{R}}_N(f)|$ for all $f \in \mathcal{F}$, such that both suprema are bounded by $\sup_{f \in \mathcal{F}} |\mathcal{R}(f) - \hat{\mathcal{R}}_N(f)|$. □

The quantity $\sup_{f \in \mathcal{F}} |\mathcal{R}(f) - \hat{\mathcal{R}}_N(f)|$ is called the **uniform deviation** between the risk and the empirical risk on $\mathcal{F}$. Note that we cannot simply consider $|\mathcal{R}(\hat{f}) - \hat{\mathcal{R}}_N(\hat{f})|$ for the single

predictor $\hat{f}$, since $\hat{f}$ depends on the data, such that $\hat{\mathcal{R}}_N(\hat{f})$ is not an average of independent random variables with expectation $\mathcal{R}(\hat{f})$ (see Section III.3). Taking the supremum over all $f \in \mathcal{F}$ removes this dependence.

INTERPRETATION AS AVERAGE OF RANDOM VARIABLES. For a fixed $f \in \mathcal{F}$, define the random variables

$$Z_n = \ell(Y_n, f(X_n)), \quad n = 1, \dots, N, \qquad \text{and} \qquad Z = \ell(Y, f(X)).$$

Since $(X_n, Y_n)_{n=1}^N$ are i.i.d. with distribution $\mathbb{P}_{X,Y}$ and f is fixed, the random variables $Z_1, \dots, Z_N$ are again i.i.d., each with the same distribution as Z. Thus

$$\hat{\mathcal{R}}_N(f) - \mathcal{R}(f) = \frac{1}{N} \sum_{n=1}^{N} Z_n - \mathbb{E}[Z],$$

i.e., bounding the uniform deviation amounts to bounding the difference of an average of i.i.d. random variables to their expectation, uniformly over all $f \in \mathcal{F}$. **Concentration inequalities** are the tool that provides such bounds: they quantify how much an average of independent random variables deviates from its expectation. In the following, we consider a single fixed f; the uniformity over $\mathcal{F}$ will be addressed later.

III.8.2 *A first concentration result: the law of large numbers*

We start with the most basic version, which shows that averaging N i.i.d. random variables reduces the variance by the factor $\frac{1}{N}$.

Proposition III.20. *Let $Z_1, \dots, Z_N$ be i.i.d. real-valued random variables with the same distribution as a random variable Z with variance $\sigma^2 = \operatorname{var}(Z) < \infty$, and let $\bar{Z}_N = \frac{1}{N} \sum_{n=1}^N Z_n$ be their average. Then*

$$\mathbb{E}[\bar{Z}_N] = \mathbb{E}[Z] \qquad \textit{and} \qquad \operatorname{var}(\bar{Z}_N) = \frac{\sigma^2}{N}.$$

Proof. By linearity of the expectation and since $\mathbb{E}[Z_n] = \mathbb{E}[Z]$ for all n,

$$\mathbb{E}[\bar{Z}_N] = \frac{1}{N} \sum_{n=1}^{N} \mathbb{E}[Z_n] = \frac{1}{N} \cdot N\mathbb{E}[Z] = \mathbb{E}[Z].$$

Using this, writing $\mu = \mathbb{E}[Z]$ and expanding the square of the sum, we obtain

$$\operatorname{var}(\bar{Z}_N) = \mathbb{E}\Big[\Big(\frac{1}{N} \sum_{n=1}^{N} (Z_n - \mu)\Big)^2\Big] = \frac{1}{N^2} \sum_{n=1}^{N} \sum_{k=1}^{N} \mathbb{E}\big[(Z_n - \mu)(Z_k - \mu)\big] = \frac{1}{N^2} \sum_{n=1}^{N} \sum_{k=1}^{N} \operatorname{cov}(Z_n, Z_k).$$

For $n \neq k$, the random variables Z_n and Z_k are independent, such that $\operatorname{cov}(Z_n, Z_k) = 0$ (see Section A.4). For $n = k$, we have $\operatorname{cov}(Z_n, Z_n) = \operatorname{var}(Z_n) = \sigma^2$. Hence only the N diagonal terms remain, and

$$\operatorname{var}(\bar{Z}_N) = \frac{1}{N^2} \sum_{n=1}^{N} \sigma^2 = \frac{1}{N^2} \cdot N\sigma^2 = \frac{\sigma^2}{N}. \qquad \square$$

Since $\mathbb{E}[\bar{Z}_N] = \mathbb{E}[Z]$, the variance of the average is exactly the expected squared difference between the average and the expectation of the underlying random variable Z:

$$\mathbb{E}\Big[\big(\bar{Z}_N - \mathbb{E}[Z]\big)^2\Big] = \operatorname{var}(\bar{Z}_N) = \frac{\sigma^2}{N}.$$

Hence, the average $\bar{Z}_N$ converges to $\mathbb{E}[Z]$ in mean square as $N \to \infty$; this is a version of the **law of large numbers**. Applied to $Z_n = \ell(Y_n, f(X_n))$, it shows that $\mathbb{E}[(\hat{\mathcal{R}}_N(f) - \mathcal{R}(f))^2] = \frac{1}{N}\operatorname{var}(\ell(Y, f(X)))$, which is already a first step towards bounds on the deviation $|\hat{\mathcal{R}}_N(f) - \mathcal{R}(f)|$, albeit in expectation. Note that the typical magnitude of the deviation itself (not squared) is $\sigma/\sqrt{N}$.

III.8.3 *From bounds in expectation to bounds in high probability*

The bound above holds in expectation, i.e., on average over all possible data sets. In practice, we have one data set and would like a statement that holds with high probability. Such a statement can always be obtained from a bound in expectation using Markov's inequality.

Lemma III.21 (Markov's inequality)**.** *Let Y be a nonnegative random variable with finite expectation. Then, for all $\varepsilon > 0$,*

$$\mathbb{P}(Y \geq \varepsilon) \leq \frac{\mathbb{E}[Y]}{\varepsilon}.$$

Proof. Since $Y \geq 0$, we have $\varepsilon \mathbb{1}\{Y \geq \varepsilon\} \leq Y$: If $Y \geq \varepsilon$, the left-hand side equals $\varepsilon \leq Y$, and otherwise it equals $0 \leq Y$. Taking expectations and using $\mathbb{E}[\mathbb{1}\{Y \geq \varepsilon\}] = \mathbb{P}(Y \geq \varepsilon)$ yields $\varepsilon \mathbb{P}(Y \geq \varepsilon) \leq \mathbb{E}[Y]$. □

Proposition III.22 (From expectation to high probability)**.** *Let Y be a nonnegative random variable with $\mathbb{E}[Y] \leq B$ for some $B > 0$. Then, for every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$Y < \frac{B}{\delta}.$$

Proof. By Markov's inequality with $\varepsilon = B/\delta$,

$$\mathbb{P}\Big(Y \geq \frac{B}{\delta}\Big) \leq \frac{\delta\, \mathbb{E}[Y]}{B} \leq \delta,$$

and hence $\mathbb{P}(Y < B/\delta) = 1 - \mathbb{P}(Y \geq B/\delta) \geq 1 - \delta$. □

APPLICATION TO AVERAGES (CHEBYSHEV'S INEQUALITY). Applying Lemma III.21 to $Y = (\bar{Z}_N - \mathbb{E}[Z])^2$ with ε^2 in place of ε, and using $\mathbb{E}[Y] = \sigma^2/N$ from Proposition III.20, we obtain for all $\varepsilon > 0$

$$\mathbb{P}\big(|\bar{Z}_N - \mathbb{E}[Z]| \geq \varepsilon\big) = \mathbb{P}\big((\bar{Z}_N - \mathbb{E}[Z])^2 \geq \varepsilon^2\big) \leq \frac{\sigma^2}{N\varepsilon^2}.$$

Equivalently, by Proposition III.22 with $B = \sigma^2/N$, with probability at least $1 - \delta$,

$$(\bar{Z}_N - \mathbb{E}[Z])^2 < \frac{\sigma^2}{N\delta}, \qquad \text{i.e.,} \qquad |\bar{Z}_N - \mathbb{E}[Z]| < \frac{\sigma}{\sqrt{N\delta}}.$$

For $Z_n = \ell(Y_n, f(X_n))$ and a fixed predictor f, this yields that with probability at least $1-\delta$, $|\hat{\mathcal{R}}_N(f) - \mathcal{R}(f)| < \sqrt{\operatorname{var}(\ell(Y, f(X)))/(N\delta)}$.

In the same way, any bound $\mathbb{E}\big[\sup_{f\in\mathcal{F}} |\mathcal{R}(f) - \hat{\mathcal{R}}_N(f)|\big] \le B$ on the expected uniform deviation yields, by Propositions III.19 and III.22, that with probability at least $1-\delta$,

$$\mathcal{R}(\hat{f}) - \inf_{f\in\mathcal{F}} \mathcal{R}(f) \le 2 \sup_{f\in\mathcal{F}} |\mathcal{R}(f) - \hat{\mathcal{R}}_N(f)| < \frac{2B}{\delta}.$$

III.8.4 *Hoeffding's inequality*

The bound obtained from Chebyshev's inequality depends on the probability δ via the factor $1/\sqrt{\delta}$, which becomes large quickly as $\delta \to 0$. We therefore want refined inequalities, which show that the probability of a large deviation of an average from its expectation decays exponentially, and which lead to a dependence on δ only via $\sqrt{\log(1/\delta)}$. The most basic such result is Hoeffding's inequality for bounded random variables [1, Proposition 1.2].

Proposition III.23 (Hoeffding's inequality)**.** *Let $Z_1, \dots, Z_N$ be independent random variables such that $Z_n \in [0,1]$ almost surely for all n. Then, for any $t \ge 0$,*

$$\mathbb{P}\Big(\frac{1}{N}\sum_{n=1}^{N} Z_n - \frac{1}{N}\sum_{n=1}^{N} \mathbb{E}[Z_n] \ge t\Big) \le \exp(-2Nt^2).$$

Proof. The proof is divided into two parts.

(1) Lemma: *If $Z \in [0,1]$ almost surely, then $\mathbb{E}[\exp(s(Z - \mathbb{E}[Z]))] \le \exp(s^2/8)$ for any $s \ge 0$.*

Let $\mu = \mathbb{E}[Z]$ and define the function

$$\phi(s) = \log\big(\mathbb{E}\big[e^{s(Z-\mu)}\big]\big),$$

which is often referred to as the cumulant generating function. Since Z is bounded, we can differentiate under the expectation and obtain

$$\phi'(s) = \frac{\mathbb{E}\big[(Z-\mu)e^{s(Z-\mu)}\big]}{\mathbb{E}\big[e^{s(Z-\mu)}\big]},$$
$$\phi''(s) = \frac{\mathbb{E}\big[(Z-\mu)^2 e^{s(Z-\mu)}\big]}{\mathbb{E}\big[e^{s(Z-\mu)}\big]} - \left(\frac{\mathbb{E}\big[(Z-\mu)e^{s(Z-\mu)}\big]}{\mathbb{E}\big[e^{s(Z-\mu)}\big]}\right)^2.$$

In particular, $\phi(0) = \log(1) = 0$ and $\phi'(0) = \mathbb{E}[Z-\mu] = 0$. To bound $\phi''(s)$ for fixed s, let $\tilde{Z}$ be a random variable whose density is obtained by reweighting the density p_Z of Z as

$$p_{\tilde{Z}}(z) = \frac{e^{s(z-\mu)}}{\mathbb{E}\big[e^{s(Z-\mu)}\big]}\, p_Z(z).$$

This is a valid density (it is nonnegative and integrates to one), it vanishes outside of $[0,1]$, and for any function g it holds that $\mathbb{E}[g(\tilde{Z})] = \mathbb{E}[g(Z)e^{s(Z-\mu)}]/\mathbb{E}[e^{s(Z-\mu)}]$. Hence, $\phi''(s) =$

$\mathbb{E}[(\tilde{Z}-\mu)^2]-(\mathbb{E}[\tilde{Z}-\mu])^2=\operatorname{var}(\tilde{Z})$ is the variance of a random variable with values in $[0,1]$. For any $\nu\in\mathbb{R}$, we have

$$\mathbb{E}[(\tilde{Z}-\nu)^2]=\mathbb{E}\big[(\tilde{Z}-\mathbb{E}[\tilde{Z}])^2\big]+(\mathbb{E}[\tilde{Z}]-\nu)^2\geq\operatorname{var}(\tilde{Z}),$$

since the cross term $2(\mathbb{E}[\tilde{Z}]-\nu)\mathbb{E}[\tilde{Z}-\mathbb{E}[\tilde{Z}]]$ vanishes. Choosing $\nu=\frac{1}{2}$ and using that $2\tilde{Z}-1\in[-1,1]$, we obtain

$$\phi''(s)=\operatorname{var}(\tilde{Z})\leq\mathbb{E}\big[(\tilde{Z}-\tfrac{1}{2})^2\big]=\frac{1}{4}\mathbb{E}\big[(2\tilde{Z}-1)^2\big]\leq\frac{1}{4}.$$

By Taylor's formula, for some $\xi\in[0,s]$,

$$\phi(s)=\phi(0)+\phi'(0)s+\frac{\phi''(\xi)}{2}s^2\leq 0+0+\frac{1}{8}s^2,$$

and taking the exponential yields the claim of the lemma.

(2) Main argument: Let $t\geq 0$, $\bar{Z}=\frac{1}{N}\sum_{n=1}^N Z_n$, such that $\mathbb{E}[\bar{Z}]=\frac{1}{N}\sum_{n=1}^N\mathbb{E}[Z_n]$, and $s>0$. Since the exponential is strictly increasing,

$$\mathbb{P}\big(\bar{Z}-\mathbb{E}[\bar{Z}]\geq t\big)=\mathbb{P}\Big(e^{s(\bar{Z}-\mathbb{E}[\bar{Z}])}\geq e^{st}\Big)\leq e^{-st}\,\mathbb{E}\Big[e^{s(\bar{Z}-\mathbb{E}[\bar{Z}])}\Big],$$

where we used Markov's inequality (Lemma III.21) for the nonnegative random variable $e^{s(\bar{Z}-\mathbb{E}[\bar{Z}])}$. Now

$$e^{s(\bar{Z}-\mathbb{E}[\bar{Z}])}=\exp\Big(\sum_{n=1}^N\frac{s}{N}(Z_n-\mathbb{E}[Z_n])\Big)=\prod_{n=1}^N\exp\Big(\frac{s}{N}(Z_n-\mathbb{E}[Z_n])\Big).$$

Since $Z_1,\dots,Z_N$ are independent, the expectation of this product is the product of the expectations (this follows from the factorization of the joint density). Applying the lemma with s/N in place of s to each factor, we obtain

$$\begin{aligned}\mathbb{P}\big(\bar{Z}-\mathbb{E}[\bar{Z}]\geq t\big)&\leq e^{-st}\prod_{n=1}^N\mathbb{E}\Big[\exp\Big(\frac{s}{N}(Z_n-\mathbb{E}[Z_n])\Big)\Big]\\&\leq e^{-st}\prod_{n=1}^N\exp\Big(\frac{s^2}{8N^2}\Big)=\exp\Big(-st+\frac{s^2}{8N}\Big).\end{aligned}$$

This holds for any $s>0$. The exponent $-st+\frac{s^2}{8N}$ is minimized for $s=4Nt$, which gives $-4Nt^2+2Nt^2=-2Nt^2$ and hence the claim (for $t=0$, the claim is trivial). □

Corollary III.24 (Two-sided Hoeffding's inequality)**.** *Let $Z_1,\dots,Z_N$ be independent random variables such that $Z_n\in[0,1]$ almost surely for all n. Then, for any $t\geq 0$,*

$$\mathbb{P}\Big(\Big|\frac{1}{N}\sum_{n=1}^N Z_n-\frac{1}{N}\sum_{n=1}^N\mathbb{E}[Z_n]\Big|\geq t\Big)\leq 2\exp(-2Nt^2).$$

Proof. With $\bar{Z}$ as above, the event $\{|\bar{Z}-\mathbb{E}[\bar{Z}]|\geq t\}$ is the union of $\{\bar{Z}-\mathbb{E}[\bar{Z}]\geq t\}$ and $\{\bar{Z}-\mathbb{E}[\bar{Z}]\leq -t\}$. The probability of the first event is bounded by $\exp(-2Nt^2)$ by Proposition III.23.

For the second event, note that the random variables $1 - Z_1, \dots, 1 - Z_N$ are again independent with values in $[0,1]$, and that

$$\bar{Z} - \mathbb{E}[\bar{Z}] \leq -t \quad \Leftrightarrow \quad \frac{1}{N}\sum_{n=1}^{N}(1 - Z_n) - \frac{1}{N}\sum_{n=1}^{N}\mathbb{E}[1 - Z_n] \geq t,$$

such that Proposition III.23, applied to the random variables $1 - Z_n$, also bounds its probability by $\exp(-2Nt^2)$. Using the union bound $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$, which follows from $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$, we obtain the claim. □

Corollary III.25 (Hoeffding's inequality on $[a,b]$)**.** *Let $a < b$ and let $Z_1, \dots, Z_N$ be independent random variables such that $Z_n \in [a,b]$ almost surely for all n, and let $\bar{Z} = \frac{1}{N}\sum_{n=1}^{N} Z_n$. Then, for any $t \geq 0$,*

$$\mathbb{P}\big(\bar{Z} - \mathbb{E}[\bar{Z}] \geq t\big) \leq \exp\Big(-\frac{2Nt^2}{(b-a)^2}\Big) \qquad \textit{and} \qquad \mathbb{P}\big(|\bar{Z} - \mathbb{E}[\bar{Z}]| \geq t\big) \leq 2\exp\Big(-\frac{2Nt^2}{(b-a)^2}\Big).$$

Proof. The random variables $W_n = (Z_n - a)/(b-a)$ are independent with values in $[0,1]$, and with $\bar{W} = \frac{1}{N}\sum_{n=1}^{N} W_n$, linearity of the expectation yields $\bar{W} - \mathbb{E}[\bar{W}] = (\bar{Z} - \mathbb{E}[\bar{Z}])/(b-a)$. Hence, the event $\{\bar{Z} - \mathbb{E}[\bar{Z}] \geq t\}$ coincides with $\{\bar{W} - \mathbb{E}[\bar{W}] \geq t/(b-a)\}$, and Proposition III.23 with $t/(b-a)$ in place of t yields the first inequality. The second follows in the same way from Corollary III.24. □

BOUNDS IN HIGH PROBABILITY. Such inequalities are typically used "in the other direction": Instead of fixing the deviation t and bounding the probability, we fix the probability $\delta \in (0,1)$ and determine the corresponding deviation. In the setting of Corollary III.25, we choose t such that the right-hand side of the two-sided inequality equals δ:

$$2\exp\Big(-\frac{2Nt^2}{(b-a)^2}\Big) = \delta \quad \Leftrightarrow \quad \frac{2Nt^2}{(b-a)^2} = \log\frac{2}{\delta} \quad \Leftrightarrow \quad t = \frac{b-a}{\sqrt{2N}}\sqrt{\log\frac{2}{\delta}}.$$

For this t, the probability that $|\bar{Z} - \mathbb{E}[\bar{Z}]| \geq t$ is at most δ. Hence, with probability at least $1 - \delta$,

$$\Big|\frac{1}{N}\sum_{n=1}^{N} Z_n - \frac{1}{N}\sum_{n=1}^{N}\mathbb{E}[Z_n]\Big| < \frac{b-a}{\sqrt{2N}}\sqrt{\log\frac{2}{\delta}}.$$

In the same way, the one-sided inequality yields that with probability at least $1 - \delta$,

$$\frac{1}{N}\sum_{n=1}^{N}\mathbb{E}[Z_n] - \frac{1}{N}\sum_{n=1}^{N} Z_n < \frac{b-a}{\sqrt{2N}}\sqrt{\log\frac{1}{\delta}},$$

where we applied the one-sided inequality to $-Z_1, \dots, -Z_N \in [-b,-a]$. Compared to the bound $\sigma/\sqrt{N\delta}$ obtained from Chebyshev's inequality, the dependence on N is again $1/\sqrt{N}$, but the dependence on δ is now only logarithmic.

APPLICATION TO BOUNDED LOSSES. Assume that the loss is bounded, i.e., $\ell(y,z) \in [0,\ell_\infty]$ for all $y, z \in \mathcal{Y}$ and some $\ell_\infty > 0$; for instance, $\ell_\infty = 1$ for the 0–1 loss. For a fixed predictor f, the random variables $Z_n = \ell(Y_n, f(X_n))$, $n = 1, \dots, N$, are i.i.d. with values in $[0,\ell_\infty]$ and

expectation $\mathbb{E}[Z_n] = \mathcal{R}(f)$ (see Section III.8.1), and their average is $\hat{\mathcal{R}}_N(f)$. Applying the above bounds with $a = 0$ and $b = \ell_\infty$ yields that, for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\left|\hat{\mathcal{R}}_N(f) - \mathcal{R}(f)\right| < \frac{\ell_\infty}{\sqrt{2N}} \sqrt{\log \frac{2}{\delta}},$$

and, with probability at least $1 - \delta$,

$$\mathcal{R}(f) - \hat{\mathcal{R}}_N(f) < \frac{\ell_\infty}{\sqrt{2N}} \sqrt{\log \frac{1}{\delta}}.$$

Note that this holds for a single predictor f that does not depend on the data, and hence does not yet apply to the learned predictor $\hat{f}$.

A

RECAP: BASIC CONCEPTS FROM PROBABILITY THEORY

In this appendix, we recall some basic concepts from probability theory that are important for this lecture. It provides details and examples for the notation introduced in Chapter II.

A.1 RANDOM VARIABLES AND THEIR DISTRIBUTION

In a probabilistic treatment of data, all random quantities arise from a probability space.

Definition A.1. *A **probability space** is a triple $(\Omega, \Sigma, \mathbb{P})$ with Ω a set (the sample space), Σ a σ-algebra of subsets of Ω, and $\mathbb{P} : \Sigma \to [0, 1]$ a probability measure. Here, a **σ-algebra** is a collection of subsets of Ω with the following properties:*

1. ***Containment:** The empty set $\emptyset$ and the sample space Ω are in Σ.*
2. ***Closure under complementation:** If $A \in \Sigma$, then its complement $A^c = \Omega \setminus A$ is also in Σ.*
3. ***Closure under countable unions:** If $\{A_i\}_{i=1}^{\infty}$ is a sequence of sets in Σ, then their union $\bigcup_{i=1}^{\infty} A_i$ is also in Σ.*

*A **probability measure** $\mathbb{P}$ is a set function $\mathbb{P} : \Sigma \to [0, 1]$ that assigns a real number to each set in the σ-algebra, satisfying:*

1. ***Non-negativity:** $\mathbb{P}(A) \geq 0$ for all $A \in \Sigma$.*
2. ***Normalization:** The total probability of the sample space is one, $\mathbb{P}(\Omega) = 1$.*
3. ***Countable additivity:** For any sequence of pairwise disjoint sets $A_1, A_2, \dots \in \Sigma$:*

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$$

*Elements of Σ are called **events**.*

In practice, we never deal with the underlying probability space $(\Omega, \Sigma, \mathbb{P})$, but rather with quantities derived from it. Those quantities are called random variables.

Definition A.2. *Let $\mathcal{X}$ be a set equipped with a σ-algebra $\Sigma_{\mathcal{X}}$. A **random variable** X with values in $\mathcal{X}$ is a function*

$$X : \Omega \to \mathcal{X}$$

such that for every set $B \in \Sigma_{\mathcal{X}}$, the preimage of B under X is an element of Σ:

$$X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \Sigma.$$

*The **distribution** (or **law**) of a random variable X, denoted $\mathbb{P}_X$, is the "push-forward" measure on $(\mathcal{X}, \Sigma_{\mathcal{X}})$ defined by*

$$\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) \quad \text{for all } B \in \Sigma_{\mathcal{X}}.$$

It is again a probability measure, now on $\mathcal{X}$. We write $X \sim \mathbb{P}_X$.

Remark A.3 (Notation for events)**.** *We use the following standard shorthand notation: For $B \in \Sigma_{\mathcal{X}}$,*

$$\{X \in B\} := X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\}, \qquad \mathbb{P}(X \in B) := \mathbb{P}(\{X \in B\}) = \mathbb{P}_X(B).$$

*Accordingly, for real-valued X we write, e.g., $\{X \leq r\}$, $\{X = x\}$ and $\mathbb{P}(X \leq r)$, $\mathbb{P}(X = x)$. If X and Y are two random variables on the same probability space with values in $\mathcal{X}$ and $\mathcal{Y}$, respectively, then also $(X, Y) : \Omega \to \mathcal{X} \times \mathcal{Y}$ is a random variable, whose distribution $\mathbb{P}_{X,Y}$ is called the **joint distribution** of X and Y. We write*

$$\mathbb{P}(X \in A, Y \in B) := \mathbb{P}(\{X \in A\} \cap \{Y \in B\}) = \mathbb{P}_{X,Y}(A \times B).$$

Remark A.4. *In practice, for this lecture, we will mostly deal with X and its distribution $\mathbb{P}_X$, and can ignore aspects of measurability, such that we can just think of:*

- *$X \in \mathcal{X}$ being a random quantity in the set $\mathcal{X}$,*
- *the distribution $\mathbb{P}_X$ being a function that maps subsets of $\mathcal{X}$ to values in $[0, 1]$, such that $\mathbb{P}_X(A) = \mathbb{P}(X \in A)$ is the probability of X being in A.*

Working with distributions directly is cumbersome, since $\mathbb{P}_X$ acts on subsets of $\mathcal{X}$ rather than on points $x \in \mathcal{X}$. In many situations of interest, however, the distribution $\mathbb{P}_X$ can be described by a function p_X that acts on points, a so-called density. In the discrete case, this is achieved via the identity

$$\mathbb{P}_X(A) = \sum_{x \in A} p_X(x),$$

and in the continuous case, the sum is replaced by an integral. To be more precise, we consider the two cases separately:

Definition A.5 (Probability density function, discrete setting)**.** *If X takes values in a countable or finite set $\mathcal{X} = \{x_1, x_2, \dots\}$, we describe its distribution using a **probability density function (PDF)** $p_X : \mathcal{X} \to [0, 1]$ (in the discrete case also often called **probability mass function (PMF)**), given as $p_X(x) = \mathbb{P}(X = x)$. Then, the probability of any event $A \subseteq \mathcal{X}$ is given by*

$$\mathbb{P}_X(A) = \sum_{x \in A} p_X(x).$$

The PDF satisfies:

1. *$0 \leq p_X(x) \leq 1$ for all $x \in \mathcal{X}$.*
2. *$\sum_{x \in \mathcal{X}} p_X(x) = 1$.*

Definition A.6 (Probability density function, continuous setting)**.** *If X takes values in $\mathcal{X} = \mathbb{R}^d$, we say X has a **probability density function (PDF)** $p_X : \mathbb{R}^d \to [0, \infty)$ (with respect to the Lebesgue measure) if, for every measurable set $B \in \Sigma_{\mathcal{X}}$, the probability is given by the integral*

$$\mathbb{P}_X(B) = \int_B p_X(x)\, dx.$$

For p_X to be a valid density, it must satisfy:

1. *$p_X(x) \geq 0$ for almost all $x \in \mathbb{R}^d$.*
2. *$\int_{\mathbb{R}^d} p_X(x)\, dx = 1$.*

Remark A.7 (Not every distribution has a density)**.** *In contrast to the discrete case, in the continuous case a density need not exist. Two examples:*

- *Let $X \in \mathbb{R}$ be 0 with probability $1/2$ and uniformly distributed on $[0, 1]$ otherwise. Then $\mathbb{P}_X(\{0\}) \geq 1/2$, but $\int_{\{0\}} p(x)\, dx = 0$ for any function p. Hence X has no density, and since X is not discrete either, it is covered by neither of the definitions above.*
- *Let U be uniformly distributed on $[0, 1]$ and $X = (U, U) \in \mathbb{R}^2$. Then X lies on the diagonal $\{(u, u) : u \in \mathbb{R}\}$ with probability one, but the diagonal has Lebesgue measure zero, such that again no density can exist. The same happens for any pair $(U, g(U))$ with g a function, e.g., an input and a deterministic label.*

*Throughout the lecture, whenever we use densities, we **assume** that they exist.*

In supervised learning, we deal with pairs (X, Y) of an input X and a label Y. In classification, the input is typically continuous ($X \in \mathbb{R}^d$), while the label is discrete ($Y \in \mathcal{Y} = \{1, \ldots, K\}$). This mixed situation is covered by the following definition.

Definition A.8 (Joint density, mixed setting)**.** *Let $X \in \mathbb{R}^d$ and $Y \in \mathcal{Y}$ with $\mathcal{Y}$ countable. A function $p_{X,Y} : \mathbb{R}^d \times \mathcal{Y} \to [0, \infty)$ is called a **joint density** of (X, Y) if*

$$\mathbb{P}(X \in A, Y \in B) = \sum_{y \in B} \int_A p_{X,Y}(x, y)\, dx \quad \text{for all measurable } A \subseteq \mathbb{R}^d,\ B \subseteq \mathcal{Y}.$$

Remark A.9 (Convention)**.** *Most of the statements below are formulated for the discrete or the continuous setting. The corresponding statement in the other setting, and in the mixed setting of Definition A.8, is obtained by replacing sums with integrals over the continuous variables, and vice versa.*

To illustrate the different situations, we consider two examples that will be used throughout this chapter.

Example A.10 (Two fair dice (discrete setting))**.** *Let $\Omega = \{1, \ldots, 6\} \times \{1, \ldots, 6\}$ be the set of 36 possible outcomes of throwing two dice. Since the dice are fair, the probability of any single outcome $\omega \in \Omega$ is $\mathbb{P}(\{\omega\}) = \frac{1}{36}$.*

Define the random variable $X : \Omega \to \{2, 3, \ldots, 12\}$ as the sum of the two faces: $X(d_1, d_2) = d_1 + d_2$.

- ***Distribution** $\mathbb{P}_X$**:** The probability of the sum being exactly k is found by counting the preimages in Ω. For $k = 4$, the event is $\{X = 4\} = \{(1, 3), (2, 2), (3, 1)\}$. Thus:*

$$\mathbb{P}_X(\{4\}) = \mathbb{P}(X = 4) = \mathbb{P}(\{(1, 3), (2, 2), (3, 1)\}) = \frac{3}{36}.$$

- ***Probability density function*** p_X***:*** *The PMF captures the "triangular" nature of the sums:*

$$p_X(x) = \frac{6 - |7 - x|}{36} \quad \text{for } x \in \{2, \dots, 12\}.$$

Example A.11 (The dart board (continuous setting))**.** *Let* $\Omega = \{(u, v) \in \mathbb{R}^2 : u^2 + v^2 \leq 1\}$ *be the unit disc, representing a dart board. Assuming a uniform throw, the probability* $\mathbb{P}$ *of hitting a region* $A \subseteq \Omega$ *is the ratio of areas:*

$$\mathbb{P}(A) = \frac{\mathit{Area}(A)}{\mathit{Area}(\Omega)} = \frac{\mathit{Area}(A)}{\pi}.$$

Define the random variable $X : \Omega \to [0, 1]$ *as the distance from the center:* $X(u, v) = \sqrt{u^2 + v^2}$*.*

- ***Distribution*** $\mathbb{P}_X$***:*** *For a distance* $r \in [0, 1]$*, the event* $\{X \leq r\}$ *corresponds to the dart landing in a smaller disc of radius* r *and area* πr^2*. Thus:*

$$\mathbb{P}_X([0, r]) = \mathbb{P}(X \leq r) = \mathbb{P}(\{(u, v) : \sqrt{u^2 + v^2} \leq r\}) = \frac{\pi r^2}{\pi} = r^2.$$

- ***Density*** p_X***:*** *Since* $\mathbb{P}_X([0, r]) = \int_0^r p_X(x)\, dx = r^2$*, we differentiate with respect to* r *to find the density:*

$$p_X(x) = 2x \quad \text{for } x \in [0, 1]$$

(and $p_X(x) = 0$ *for* $x \notin [0, 1]$*).*

A.2 EXPECTATION AND MOMENTS

The moments of a distribution characterize its center and spread. The most fundamental moment is the expectation.

Definition A.12 (Expectation)**.** *The* ***expectation*** *of a random variable* X*, denoted* $\mathbb{E}[X]$*, is defined as follows:*

- ***Discrete case (*** $\mathcal{X} = \{x_1, x_2, \dots\} \subset \mathbb{R}^d$***):***

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x\, p_X(x),$$

provided that $\sum_{x \in \mathcal{X}} \|x\|\, p_X(x) < \infty$*.*

- ***Continuous case (*** $\mathcal{X} = \mathbb{R}^d$***):***

$$\mathbb{E}[X] = \int_{\mathbb{R}^d} x\, p_X(x)\, dx,$$

provided that $\int_{\mathbb{R}^d} \|x\|\, p_X(x)\, dx < \infty$*.*

If the respective condition holds, we say that X *is* ***integrable*** *or has finite expectation. For* $d > 1$*, the expectation is a vector, which is computed componentwise.*

Remark A.13. *Intuitively,* $\mathbb{E}[X]$ *is the long-run average value of* X *over many independent repetitions of the same experiment. This is made precise by the law of large numbers.*

Often, we are interested in the expectation of a function of a random variable, e.g., of the loss $\ell(Y, f(X))$ of a predictor f. Since $g(X)$ is again a random variable, its expectation could in principle be computed from its own density $p_{g(X)}$. The following result shows that this is not necessary.

Proposition A.14 (Expectation of functions of random variables)**.** *Let X be a random variable with values in $\mathcal{X}$ and $g : \mathcal{X} \to \mathbb{R}^m$ a function such that $g(X)$ is integrable. Then*

$$\mathbb{E}[g(X)] = \sum_{x \in \mathcal{X}} g(x)\, p_X(x) \quad \textit{(discrete case)}, \qquad \mathbb{E}[g(X)] = \int_{\mathcal{X}} g(x)\, p_X(x)\, dx \quad \textit{(continuous case)}.$$

In particular, for two random variables X, Y and $h : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^m$, it holds that $\mathbb{E}[h(X,Y)] = \int_{\mathcal{X}} \int_{\mathcal{Y}} h(x,y)\, p_{X,Y}(x,y)\, dy\, dx$, with integrals replaced by sums for discrete variables.

Proof. We only consider the discrete case. With $\mathcal{Z} = g(\mathcal{X})$ the (countable) set of values of $Z = g(X)$, we have

$$\begin{aligned}\mathbb{E}[g(X)] &= \sum_{z \in \mathcal{Z}} z\, \mathbb{P}(g(X) = z) = \sum_{z \in \mathcal{Z}} z \sum_{x \in \mathcal{X} : g(x) = z} p_X(x) \\ &= \sum_{z \in \mathcal{Z}} \sum_{x \in \mathcal{X} : g(x) = z} g(x) p_X(x) = \sum_{x \in \mathcal{X}} g(x) p_X(x),\end{aligned}$$

where rearranging the sum is allowed due to absolute convergence. □

Remark A.15 (Important consequences)**.**

- ***Linearity:*** *For integrable X, Y and $a, b \in \mathbb{R}$, it holds that $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ (choose $h(x,y) = ax + by$ above).*
- ***Probabilities as expectations:*** *For $A \subseteq \mathcal{X}$ and the indicator function $\mathbb{1}_A$ (with $\mathbb{1}_A(x) = 1$ if $x \in A$ and $\mathbb{1}_A(x) = 0$ else), it holds that*

$$\mathbb{E}[\mathbb{1}_A(X)] = \int_{\mathcal{X}} \mathbb{1}_A(x) p_X(x)\, dx = \int_A p_X(x)\, dx = \mathbb{P}(X \in A).$$

*Both will be used throughout the lecture. For instance, the **risk** of a predictor $f : \mathcal{X} \to \mathcal{Y}$ with respect to a loss function ℓ is the expected loss $\mathbb{E}[\ell(Y, f(X))]$, and the **coverage** of a prediction set $C(X_{test})$ as considered in conformal prediction is the probability*

$$\mathbb{P}\big(Y_{test} \in C(X_{test})\big) = \mathbb{E}\big[\mathbb{1}\{Y_{test} \in C(X_{test})\}\big],$$

where we write $\mathbb{1}\{\text{statement}\}$ for the random variable that is one if the statement holds and zero otherwise.

Definition A.16 (Variance and covariance)**.** *Using expectations, we can define the spread of a random variable and relationships between random variables without distinguishing between discrete and continuous cases. Here, we assume that the random variables are real-valued, i.e., $\mathcal{X} \subseteq \mathbb{R}$, and that they have finite second moments, i.e., $\mathbb{E}[X^2] < \infty$.*

VARIANCE: *Measures the average squared deviation of X from its mean $\mu = \mathbb{E}[X]$:*

$$\operatorname{var}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

COVARIANCE: *For two random variables X_1 and X_2 defined on the same probability space, the covariance measures their joint variability:*

$$\operatorname{cov}(X_1, X_2) = \mathbb{E}\left[(X_1 - \mathbb{E}[X_1])(X_2 - \mathbb{E}[X_2])\right] = \mathbb{E}[X_1 X_2] - \mathbb{E}[X_1]\mathbb{E}[X_2].$$

*In the vector-valued case $X = (X_1, \dots, X_d) \in \mathbb{R}^d$, we define the **covariance matrix** of X as*

$$\operatorname{cov}(X) = \mathbb{E}\left[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top\right] = \left(\operatorname{cov}(X_j, X_k)\right)_{j,k=1}^{d} \in \mathbb{R}^{d\times d}.$$

Example A.17 (Two fair dice (discrete))**.** *Let X_1, X_2 denote the faces of the first and the second die, i.e., $X_1(d_1, d_2) = d_1$ and $X_2(d_1, d_2) = d_2$, such that the sum is $X = X_1 + X_2$. Both X_1 and X_2 are uniformly distributed on $\{1, \dots, 6\}$, i.e., $p_{X_i}(k) = \frac{1}{6}$ for $k \in \{1, \dots, 6\}$, $i = 1, 2$. First, we calculate the expectation of a single die:*

$$\mathbb{E}[X_i] = \sum_{k=1}^{6} k\, p_{X_i}(k) = \sum_{k=1}^{6} k \cdot \frac{1}{6} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5.$$

*By the **linearity of expectation**, the expectation of the sum is the sum of the expectations:*

$$\mathbb{E}[X] = \mathbb{E}[X_1 + X_2] = \mathbb{E}[X_1] + \mathbb{E}[X_2] = 3.5 + 3.5 = 7.$$

To compute the variance of the sum X, we use that X takes values $x \in \{2, 3, \dots, 12\}$, that $\mu = \mathbb{E}[X] = 7$, and Proposition A.14 to obtain

$$\operatorname{var}(X) = \sum_{x=2}^{12} (x-7)^2\, p_X(x) = \frac{2\cdot(25\cdot 1 + 16\cdot 2 + 9\cdot 3 + 4\cdot 4 + 1\cdot 5)}{36} = \frac{210}{36} = \frac{35}{6} \approx 5.83.$$

Example A.18 (The dart board (continuous))**.** *Recall the density for the distance X was $p_X(x) = 2x$ for $x \in [0, 1]$.*

- ***Expectation:***
$$\mathbb{E}[X] = \int_0^1 x(2x)\, dx = \int_0^1 2x^2\, dx = \left[\frac{2}{3}x^3\right]_0^1 = \frac{2}{3}.$$
The "average" distance of a dart from the center is $2/3$, not $1/2$, because there is more area (and thus more probability) near the edge of the disc.

- ***Variance:*** *First, compute $\mathbb{E}[X^2] = \int_0^1 x^2(2x)\, dx = \int_0^1 2x^3\, dx = \frac{1}{2}$. Then*
$$\operatorname{var}(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18} \approx 0.056.$$

A.3 DISTRIBUTION FUNCTIONS AND QUANTILES

For real-valued random variables, the distribution can always be described by a function on points, even if no density exists.

Definition A.19 (Cumulative distribution function)**.** *The **cumulative distribution function (CDF)** of a real-valued random variable X is the function $F_X : \mathbb{R} \to [0,1]$ given as*

$$F_X(t) = \mathbb{P}(X \leq t).$$

Remark A.20. *The CDF F_X is non-decreasing, right-continuous, and satisfies $\lim_{t\to-\infty} F_X(t) = 0$ and $\lim_{t\to\infty} F_X(t) = 1$. Moreover, $\mathbb{P}(X < t) = \lim_{s\uparrow t} F_X(s)$, such that F_X is continuous at t if and only if $\mathbb{P}(X = t) = 0$. If X has a density p_X, then $F_X(t) = \int_{-\infty}^{t} p_X(x)\,dx$.*

Quantiles invert the CDF: The β-quantile is the smallest value that X does not exceed with probability at least β.

Definition A.21 (Quantile)**.** *Let X be a real-valued random variable and $\beta \in (0,1)$. The **β-quantile** of X is*

$$q_X(\beta) = \inf\{t \in \mathbb{R} : F_X(t) \geq \beta\}.$$

*The $1/2$-quantile is called the **median**.*

Lemma A.22. *Let X be a real-valued random variable and $\beta \in (0,1)$. Then $q_X(\beta) \in \mathbb{R}$ and*

$$\mathbb{P}(X < q_X(\beta)) \leq \beta \leq \mathbb{P}(X \leq q_X(\beta)).$$

In particular, if F_X is continuous at $q_X(\beta)$, then $\mathbb{P}(X \leq q_X(\beta)) = \beta$.

Proof. Let $T = \{t \in \mathbb{R} : F_X(t) \geq \beta\}$. Since $F_X(t) \to 1 > \beta$ as $t \to \infty$, T is non-empty, and since $F_X(t) \to 0 < \beta$ as $t \to -\infty$, T is bounded from below. Hence $q = q_X(\beta) = \inf T \in \mathbb{R}$. Taking $t_k \in T$ with $t_k \downarrow q$, right-continuity yields $F_X(q) = \lim_k F_X(t_k) \geq \beta$, i.e., $\mathbb{P}(X \leq q) \geq \beta$. Further, for any $s < q$ we have $s \notin T$, i.e., $F_X(s) < \beta$, such that $\mathbb{P}(X < q) = \lim_{s\uparrow q} F_X(s) \leq \beta$. If F_X is continuous at q, then $\mathbb{P}(X < q) = \mathbb{P}(X \leq q)$, which implies the last assertion. □

Example A.23.

- ***Dart board:** For the distance X, we have $F_X(t) = t^2$ for $t \in [0,1]$ (and $F_X(t) = 0$ for $t < 0$, $F_X(t) = 1$ for $t > 1$). Hence $q_X(\beta) = \sqrt{\beta}$, and the median distance is $1/\sqrt{2} \approx 0.71$.*
- ***Two dice:** For the sum X, we have $F_X(6) = \frac{15}{36} < \frac{1}{2}$ and $F_X(7) = \frac{21}{36} \geq \frac{1}{2}$. Hence the median is $q_X(1/2) = 7$, while $\mathbb{P}(X \leq 7) = \frac{21}{36} > \frac{1}{2}$.*

In practice, the distribution of X is unknown and we only have access to samples $z_1, \ldots, z_N \in \mathbb{R}$. Replacing $\mathbb{P}_X$ by the uniform distribution on these samples leads to the empirical counterparts of the above notions.

Definition A.24 (Empirical CDF and empirical quantile)**.** *Let $z_1, \ldots, z_N \in \mathbb{R}$, and denote by $z_{(1)} \leq z_{(2)} \leq \ldots \leq z_{(N)}$ the same values sorted in increasing order (the **order statistics**). The **empirical CDF** is*

$$\hat{F}_N(t) = \frac{1}{N}\sum_{n=1}^{N} \mathbb{1}\{z_n \leq t\} = \frac{\#\{n : z_n \leq t\}}{N},$$

*and, for $\beta \in (0,1]$, the **empirical β-quantile** is $\hat{q}_N(\beta) = \inf\{t \in \mathbb{R} : \hat{F}_N(t) \geq \beta\}$. For $\beta > 1$, we use the convention $\hat{q}_N(\beta) = \infty$.*

Lemma A.25. *For $\beta \in (0,1]$, the empirical β-quantile is given as $\hat{q}_N(\beta) = z_{(\lceil N\beta \rceil)}$.*

Proof. Since $\#\{n : z_n \le t\}$ is an integer, we have

$$\hat{F}_N(t) \ge \beta \quad \Leftrightarrow \quad \#\{n : z_n \le t\} \ge N\beta \quad \Leftrightarrow \quad \#\{n : z_n \le t\} \ge \lceil N\beta \rceil \quad \Leftrightarrow \quad t \ge z_{(\lceil N\beta \rceil)}.$$

Hence $\{t \in \mathbb{R} : \hat{F}_N(t) \ge \beta\} = [z_{(\lceil N\beta \rceil)}, \infty)$, whose infimum is $z_{(\lceil N\beta \rceil)}$. □

Remark A.26. *In conformal prediction, we will compute scores $s_1, \dots, s_N$ on calibration data and use the empirical quantile*

$$\hat{q} = \hat{q}_N\left(\frac{\lceil (N+1)(1-\alpha) \rceil}{N}\right) = s_{(\lceil (N+1)(1-\alpha) \rceil)},$$

where $\alpha \in (0,1)$ is a user-chosen error rate. The slightly increased level $\lceil (N+1)(1-\alpha) \rceil / N \ge 1-\alpha$ is a finite-sample correction that will be explained later. If $\lceil (N+1)(1-\alpha) \rceil > N$, the convention above gives $\hat{q} = \infty$.

A.4 INDEPENDENCE, MARGINAL AND CONDITIONAL DISTRIBUTIONS

Of central importance is the notion of independence for random variables. We start with the independence of events.

Definition A.27 (Independence of events)**.** *In the probability space $(\Omega, \Sigma, \mathbb{P})$, two events $A, B \in \Sigma$ are said to be **independent** if*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Intuitively, the occurrence of one event does not change the probability of the other (see also the conditional probability below).

Example A.28 (Independence and dependence in dice rolls)**.** *Consider the sample space $\Omega = \{1, \dots, 6\}^2$ of two fair dice rolls.*

1. ***Independent events:** Let A be the event that the first die is a 3, and B be the event that the second die is a 5.*

   $$A = \{(3,1), (3,2), \dots, (3,6)\}, \quad B = \{(1,5), (2,5), \dots, (6,5)\}$$

   We have $\mathbb{P}(A) = \frac{6}{36} = \frac{1}{6}$ and $\mathbb{P}(B) = \frac{6}{36} = \frac{1}{6}$. The intersection $A \cap B$ is the single outcome $\{(3,5)\}$, so:

   $$\mathbb{P}(A \cap B) = \frac{1}{36}.$$

   *Since $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$, the events A and B are **independent**. Knowing the result of the first die tells us nothing about the second.*

2. ***Dependent (non-independent) events:** Let C be the event that the sum of the dice is 4, and D be the event that the second die is even.*

   $$C = \{(1,3), (2,2), (3,1)\}, \quad D = \{(d_1, d_2) : d_2 \in \{2,4,6\}\}$$

   We have $\mathbb{P}(C) = \frac{3}{36} = \frac{1}{12}$ and $\mathbb{P}(D) = \frac{18}{36} = \frac{1}{2}$. The intersection $C \cap D$ is the single outcome where the sum is 4 and the second die is even: $\{(2,2)\}$. Hence

   $$\mathbb{P}(C \cap D) = \frac{1}{36}.$$

*However, $\mathbb{P}(C)\mathbb{P}(D) = \frac{1}{12} \times \frac{1}{2} = \frac{1}{24}$. Since $\frac{1}{36} \neq \frac{1}{24}$, the events are **dependent**. If you know the sum is 4, the second die is less likely to be even than it would be otherwise.*

Now we consider the independence of random variables.

Definition A.29 (Independence of random variables)**.** *Two random variables $X_1 \in \mathcal{X}_1$ and $X_2 \in \mathcal{X}_2$ defined on the same probability space are **independent** if the events $\{X_1 \in B_1\}$ and $\{X_2 \in B_2\}$ are independent for all measurable sets $B_1 \subseteq \mathcal{X}_1$, $B_2 \subseteq \mathcal{X}_2$, i.e.,*

$$\mathbb{P}(X_1 \in B_1, X_2 \in B_2) = \mathbb{P}(X_1 \in B_1)\mathbb{P}(X_2 \in B_2).$$

More generally, $X_1, \ldots, X_d$ are independent if $\mathbb{P}(X_1 \in B_1, \ldots, X_d \in B_d) = \prod_{j=1}^{d} \mathbb{P}(X_j \in B_j)$ for all measurable sets $B_1, \ldots, B_d$.

If densities exist, independence can be characterized via a factorization of the joint density, which is how we will typically verify it.

Proposition A.30 (Independence via densities)**.** *Let $X_1, \ldots, X_d$ be random variables with densities $p_{X_1}, \ldots, p_{X_d}$ and joint density $p_{X_1,\ldots,X_d}$. Then $X_1, \ldots, X_d$ are independent if and only if*

$$p_{X_1,\ldots,X_d}(x_1, \ldots, x_d) = p_{X_1}(x_1) \cdot \ldots \cdot p_{X_d}(x_d)$$

for all $(x_1, \ldots, x_d)$ in the discrete case, and for almost all $(x_1, \ldots, x_d)$ in the continuous case.

Proof. We only show the case $d = 2$ in the discrete setting. If X_1, X_2 are independent, choosing $B_1 = \{x_1\}$, $B_2 = \{x_2\}$ yields $p_{X_1,X_2}(x_1, x_2) = \mathbb{P}(X_1 = x_1, X_2 = x_2) = p_{X_1}(x_1)p_{X_2}(x_2)$. Conversely, if the density factorizes, then

$$\mathbb{P}(X_1 \in B_1, X_2 \in B_2) = \sum_{x_1 \in B_1} \sum_{x_2 \in B_2} p_{X_1}(x_1)p_{X_2}(x_2) = \mathbb{P}(X_1 \in B_1)\mathbb{P}(X_2 \in B_2). \qquad \square$$

Example A.31 (Independent vs. dependent random variables)**.** *Consider again the rolling of two fair dice. Let X_1 be the outcome of the first die and X_2 be the outcome of the second.*

1. ***Independent variables:** The random variables X_1 and X_2 are independent. For any pair $(k, m) \in \{1, \ldots, 6\}^2$:*

   $$p_{X_1,X_2}(k, m) = \frac{1}{36}.$$

   Since $p_{X_1}(k) = \frac{1}{6}$ and $p_{X_2}(m) = \frac{1}{6}$, we see that $p_{X_1,X_2}(k, m) = p_{X_1}(k)p_{X_2}(m)$ holds for all k, m.

2. ***Dependent variables:** Let $S = X_1 + X_2$ be the sum of the dice. The random variables S and X_1 are **not** independent. To prove this, we only need to find one counterexample. Let $s = 12$ and $k = 1$:*

   $$p_{S,X_1}(12, 1) = 0$$

   (since if the first die is a 1, the maximum sum is 7). However:

   $$p_S(12)p_{X_1}(1) = \frac{1}{36} \cdot \frac{1}{6} = \frac{1}{216}.$$

   Since $0 \neq \frac{1}{216}$, the variables S and X_1 are dependent. Intuitively, knowing the value of the first die changes our knowledge about the possible range of the sum.

Remark A.32.

- *If two real-valued random variables X_1, X_2 with finite second moments are independent, then $\mathbb{E}[X_1 X_2] = \mathbb{E}[X_1]\mathbb{E}[X_2]$ and hence $\operatorname{cov}(X_1, X_2) = 0$.*
- *Conversely, it is important to note that $\operatorname{cov}(X_1, X_2) = 0$ does in general not imply independence. For example, if X_1 is uniformly distributed on $\{-1, 0, 1\}$ and $X_2 = X_1^2$, then $\operatorname{cov}(X_1, X_2) = \mathbb{E}[X_1^3] - \mathbb{E}[X_1]\mathbb{E}[X_1^2] = 0$, but X_2 is a function of X_1. In some special cases, e.g., if X_1, X_2 are jointly Gaussian, uncorrelated random variables are independent.*

When dealing with multiple random variables, we often need to isolate the behavior of a single variable, regardless of the outcomes of the others. This leads to the concept of the marginal distribution.

Definition A.33 (Marginal distribution)**.** *Let (X_1, X_2) be a joint random variable taking values in $\mathcal{X}_1 \times \mathcal{X}_2$ with joint distribution $\mathbb{P}_{X_1,X_2}$. The **marginal distribution** of X_1, denoted $\mathbb{P}_{X_1}$, is obtained by evaluating the joint distribution $\mathbb{P}_{X_1,X_2}$ over the entire space of the second variable:*

$$\mathbb{P}_{X_1}(A) = \mathbb{P}_{X_1,X_2}(A \times \mathcal{X}_2) \quad \text{for any measurable set } A \subseteq \mathcal{X}_1.$$

Intuitively, $\mathbb{P}_{X_1}(A)$ is the probability that $X_1 \in A$ occurs, allowing X_2 to take any possible value in $\mathcal{X}_2$.

Remark A.34 (Marginal densities)**.** *In practice, we compute the density of the marginal distribution by "summing out" or "integrating out" the unwanted variable from the joint density p_{X_1,X_2}.*

- ***Discrete case:** If X_1 and X_2 are discrete, the marginal probability density function is obtained by summing over all possible values of X_2:*

$$p_{X_1}(x_1) = \sum_{x_2 \in \mathcal{X}_2} p_{X_1,X_2}(x_1, x_2).$$

- ***Continuous case:** If X_1 and X_2 are continuous, the marginal probability density function is obtained by integrating over the entire range of X_2:*

$$p_{X_1}(x_1) = \int_{\mathcal{X}_2} p_{X_1,X_2}(x_1, x_2)\, dx_2.$$

- ***Mixed case:** For $X \in \mathbb{R}^d$ continuous and $Y \in \mathcal{Y}$ discrete with joint density $p_{X,Y}$ as in Definition A.8, we have*

$$p_X(x) = \sum_{y \in \mathcal{Y}} p_{X,Y}(x, y), \qquad p_Y(y) = \mathbb{P}(Y = y) = \int_{\mathbb{R}^d} p_{X,Y}(x, y)\, dx.$$

*This process is often called **marginalization**.*

Example A.35 (Marginal distribution for the dart board)**.** *Let the sample space Ω be the unit disc $D = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 1\}$, and let X_1, X_2 be the two coordinates of the point that is hit. If the dart hits are uniform over the disc, the joint density $p_{X_1,X_2}(x_1, x_2)$ is constant on D:*

$$p_{X_1,X_2}(x_1, x_2) = \begin{cases} \frac{1}{\pi} & \text{if } x_1^2 + x_2^2 \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

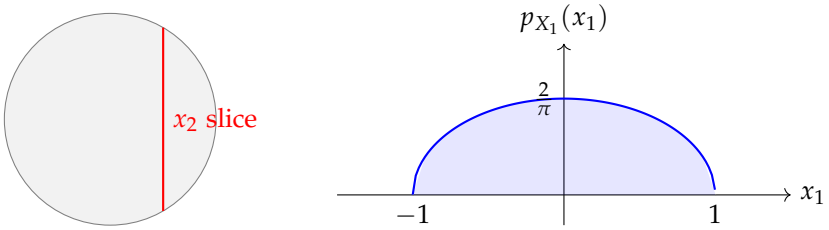


Figure A.1: Visualization of marginal distribution for the dart example: Left: The dart disc with a slice corresponding to a a fixed value of X_1. Right: The marginal distribution of X_1.

To find the marginal density $p_{X_1}(x_1)$, we integrate out x_2. For a fixed $x_1 \in [-1, 1]$, the variable x_2 ranges from $-\sqrt{1-x_1^2}$ to $\sqrt{1-x_1^2}$:

$$p_{X_1}(x_1) = \int_{-\infty}^{\infty} p_{X_1,X_2}(x_1, x_2)\, dx_2 = \int_{-\sqrt{1-x_1^2}}^{\sqrt{1-x_1^2}} \frac{1}{\pi}\, dx_2.$$

Evaluating the integral:

$$p_{X_1}(x_1) = \frac{1}{\pi}\,[x_2]_{-\sqrt{1-x_1^2}}^{\sqrt{1-x_1^2}} = \frac{2\sqrt{1-x_1^2}}{\pi} \quad \text{for } x_1 \in [-1, 1].$$

*Notice that $p_{X_1}(x_1)$ is **not uniform**. It follows a semi-elliptical shape, reflecting the fact that there is "more area" to hit near the center ($x_1 = 0$) than near the edges ($x_1 \approx \pm 1$). See also Fig. A.1 for a visualization.*

Finally, we also need the notions of conditional probability and conditional distribution.

Definition A.36 (Conditional probability of events)**.** *In a probability space $(\Omega, \Sigma, \mathbb{P})$, for two events $A, B \in \Sigma$ with $\mathbb{P}(B) > 0$, the **conditional probability** of A given B is defined as*

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

This represents the probability that event A occurs, restricted to the reduced sample space where B is known to have occurred.

Note that two events A, B with $\mathbb{P}(B) > 0$ are independent if and only if $\mathbb{P}(A|B) = \mathbb{P}(A)$.

Definition A.37 (Conditional density)**.** *The **conditional density** of a random variable X_1 given $X_2 = x_2$ is defined by the ratio of the joint density to the marginal density:*

$$p_{X_1|X_2}(x_1|x_2) = \frac{p_{X_1,X_2}(x_1, x_2)}{p_{X_2}(x_2)}.$$

This is well-defined for any x_2 such that $p_{X_2}(x_2) > 0$. For fixed x_2, the function $x_1 \mapsto p_{X_1|X_2}(x_1|x_2)$ is again a probability density.

Remark A.38. *Note that here we rely on densities to introduce the notion of conditioning for random variables. From the conditional density, also the* ***conditional distribution*** *of X_1 given $X_2 = x_2$ can be obtained via*

$$\mathbb{P}_{X_1|X_2=x_2}(A) = \mathbb{P}(X_1 \in A | X_2 = x_2) = \int_A p_{X_1|X_2}(x_1|x_2)\, dx_1.$$

In the discrete case, this coincides with the conditional probability of the event $\{X_1 \in A\}$ given $\{X_2 = x_2\}$. In the continuous case, $\mathbb{P}(X_2 = x_2) = 0$, such that the definition of conditional probability for events is not applicable, and the definition via densities is necessary. The conditional distribution can also be defined without using densities, but this requires advanced measure theory and goes beyond the scope of this lecture.

Remark A.39 (Relation to independence)**.** *From the factorization of the joint density ($p_{X_1,X_2} = p_{X_1} p_{X_2}$), it follows that if X_1 and X_2 are independent, then for all x_2 with $p_{X_2}(x_2) > 0$:*

$$p_{X_1|X_2}(x_1|x_2) = \frac{p_{X_1}(x_1) p_{X_2}(x_2)}{p_{X_2}(x_2)} = p_{X_1}(x_1).$$

In other words, for independent variables, the conditional distribution is simply the marginal distribution; knowing X_2 provides no additional information about X_1.

Example A.40 (Conditional dice sums)**.** *Let X_1 be the result of the first die and $S = X_1 + X_2$ be the sum of two dice. We previously saw that these are dependent. Let us compute the conditional distribution of the sum S given that we know the first die is $X_1 = 2$.*

- ***Marginal:*** *$p_{X_1}(2) = \frac{1}{6}$.*
- ***Joint:*** *The event $\{S = s, X_1 = 2\}$ is only possible if $s = 2 + k$ where $k \in \{1, \dots, 6\}$. Thus, $p_{S,X_1}(s, 2) = \frac{1}{36}$ for $s \in \{3, 4, 5, 6, 7, 8\}$, and 0 otherwise.*
- ***Conditional:*** *Applying the definition:*

$$p_{S|X_1}(s|2) = \frac{p_{S,X_1}(s,2)}{p_{X_1}(2)} = \frac{1/36}{1/6} = \frac{1}{6} \quad \text{for } s \in \{3, \dots, 8\},$$

and $p_{S|X_1}(s|2) = 0$ else.

While the marginal distribution of S is "triangular" (peaking at 7), the ***conditional*** *distribution $\mathbb{P}_{S|X_1=2}$ is* ***uniform*** *on $\{3, \dots, 8\}$. Knowing X_1 fundamentally reshapes our distribution of S.*

An important relation between conditional probabilities, and between conditional densities, is provided by Bayes' theorem.

Proposition A.41 (Bayes' theorem)**.** *For two events A, B in a probability space $(\Omega, \Sigma, \mathbb{P})$ with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$ it holds that*

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Further, for two random variables $X \in \mathcal{X}$ and $Y \in \mathcal{Y}$ with joint density $p_{X,Y}$ it holds that

$$p_{X|Y}(x|y) = \frac{p_{Y|X}(y|x) p_X(x)}{p_Y(y)}$$

for all x, y with $p_X(x) > 0$ and $p_Y(y) > 0$.

Proof. We only provide the proof for the statement with densities; the one for events works the same. By definition of the conditional density, for $p_X(x) > 0$ and $p_Y(y) > 0$ we have

$$p_{X|Y}(x|y)\, p_Y(y) = p_{X,Y}(x,y) = p_{Y|X}(y|x)\, p_X(x).$$

Dividing by $p_Y(y)$ yields the claim. □

Based on conditional densities, we can also define the conditional expectation.

Definition A.42 (Conditional expectation)**.** *Let X and Y be random variables and $h : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}^m$ such that $h(X,Y)$ is integrable. For y with $p_Y(y) > 0$, the **conditional expectation** of $h(X,Y)$ given $Y = y$, denoted $\mathbb{E}[h(X,Y)|Y = y]$, is the mean with respect to the conditional distribution of X given $Y = y$:*

- ***Continuous case:***
$$\mathbb{E}[h(X,Y)|Y = y] = \int_{\mathcal{X}} h(x,y)\, p_{X|Y}(x|y)\, dx.$$
- ***Discrete case:***
$$\mathbb{E}[h(X,Y)|Y = y] = \sum_{x \in \mathcal{X}} h(x,y)\, p_{X|Y}(x|y).$$

The most important special case is $h(x,y) = x$, which yields $\mathbb{E}[X|Y = y]$, the mean of the conditional distribution of X given $Y = y$.

Remark A.43 (Conditional expectation as a random variable)**.** *It is crucial to distinguish between two notations:*

1. *$\mathbb{E}[X|Y = y]$ is a **numerical value** (a constant) for a specific value y.*
2. *$\mathbb{E}[X|Y]$ is a **random variable** that is a function of Y. Its value is $g(Y)$, where $g(y) = \mathbb{E}[X|Y = y]$.*

Proposition A.44 (Rules for conditional expectations)**.** *Let X, Y be random variables, $a, b \in \mathbb{R}$, and assume all occurring expectations exist. Then, for all y with $p_Y(y) > 0$:*

1. ***Linearity:** $\mathbb{E}[aX_1 + bX_2|Y = y] = a\mathbb{E}[X_1|Y = y] + b\mathbb{E}[X_2|Y = y]$ for random variables X_1, X_2 with values in $\mathbb{R}^m$.*
2. ***Taking out what is known:** For $g : \mathcal{Y} \to \mathbb{R}$, it holds that $\mathbb{E}[g(Y)X|Y = y] = g(y)\mathbb{E}[X|Y = y]$, i.e., $\mathbb{E}[g(Y)X|Y] = g(Y)\mathbb{E}[X|Y]$. In particular, $\mathbb{E}[g(Y)|Y = y] = g(y)$ and $\mathbb{E}[Y|Y = y] = y$.*
3. ***Independence:** If X and Y are independent, then $\mathbb{E}[X|Y = y] = \mathbb{E}[X]$.*

Proof. We consider the continuous case. Linearity follows from the linearity of the integral (with $h(x,y) = ax_1 + bx_2$ and $X = (X_1, X_2)$). For the second assertion, the definition with $h(x,y) = g(y)x$ yields

$$\mathbb{E}[g(Y)X|Y = y] = \int_{\mathcal{X}} g(y)\, x\, p_{X|Y}(x|y)\, dx = g(y) \int_{\mathcal{X}} x\, p_{X|Y}(x|y)\, dx = g(y)\mathbb{E}[X|Y = y].$$

Choosing $h(x,y) = g(y)$ yields $\mathbb{E}[g(Y)|Y = y] = g(y) \int_{\mathcal{X}} p_{X|Y}(x|y)\, dx = g(y)$. The last assertion follows from $p_{X|Y}(x|y) = p_X(x)$ in case of independence. □

Finally, we recall the **law of total expectation** (also known as the **tower property**), which is a fundamental tool in statistical learning theory. It allows us to break down the expectation of a random variable X by first conditioning on a related variable Y (like the input of a predictor, or a class label) and then averaging over the distribution of that related variable.

Proposition A.45 (Law of total expectation)**.** *Let X and Y be random variables defined on the same probability space and X integrable. The expectation of X can be computed by taking the expectation of the conditional expectation of X given Y:*

$$\mathbb{E}[X] = \mathbb{E}_Y[\mathbb{E}_{X|Y}[X|Y]].$$

In the continuous case, this expands to

$$\mathbb{E}[X] = \int_{\mathcal{Y}} \left(\int_{\mathcal{X}} x \, p_{X|Y}(x|y) \, dx \right) p_Y(y) \, dy,$$

and in the discrete case to

$$\mathbb{E}[X] = \sum_{y \in \mathcal{Y}} \mathbb{E}[X|Y = y] \, p_Y(y).$$

More generally, $\mathbb{E}[h(X,Y)] = \mathbb{E}_Y[\mathbb{E}[h(X,Y)|Y]]$ for any h such that $h(X,Y)$ is integrable.

Proof. We consider the continuous case and $h(X,Y)$. Let $\mathcal{Y}_+ = \{y \in \mathcal{Y} : p_Y(y) > 0\}$. Since $p_{X,Y}(x,y) = 0$ for almost all (x,y) with $y \notin \mathcal{Y}_+$ (as $p_Y(y) = \int_{\mathcal{X}} p_{X,Y}(x,y)\,dx = 0$ for those y), using $p_{X|Y}(x|y)p_Y(y) = p_{X,Y}(x,y)$ for $y \in \mathcal{Y}_+$, Fubini's theorem and Proposition A.14, we obtain

$$\int_{\mathcal{Y}_+} \left(\int_{\mathcal{X}} h(x,y) \, p_{X|Y}(x|y) \, dx \right) p_Y(y) \, dy = \int_{\mathcal{Y}} \int_{\mathcal{X}} h(x,y) \, p_{X,Y}(x,y) \, dx \, dy = \mathbb{E}[h(X,Y)]. \qquad \square$$

EXPLANATION AND INTUITION. The law of total expectation tells us that to find the "global" average of X, we can:

1. Find the "local" average of X for every possible fixed value of Y (this is the inner term $\mathbb{E}[X|Y]$).
2. Average these local means together, weighting each one by how likely that value of Y is to occur (this is the outer expectation $\mathbb{E}_Y[\dots]$).

In statistical learning, we will frequently apply this to the risk of a predictor f, conditioning on the input:

$$\mathbb{E}[\ell(Y, f(X))] = \mathbb{E}_X\big[\mathbb{E}[\ell(Y, f(X))|X]\big].$$

This allows us to analyze the risk pointwise for each fixed input x.

Example A.46 (Application to dice)**.** *Recall our sum $S = X_1 + X_2$. Suppose we want to compute $\mathbb{E}[S]$ by conditioning on the first die X_1.*

- *Using Proposition A.44 and the independence of X_1 and X_2, we obtain $\mathbb{E}[S|X_1 = k] = \mathbb{E}[X_1 + X_2|X_1 = k] = \mathbb{E}[X_1|X_1 = k] + \mathbb{E}[X_2|X_1 = k] = k + \mathbb{E}[X_2] = k + 3.5$.*
- *Now we take the expectation of this result over all possible $k \in \{1, \dots, 6\}$:*

$$\mathbb{E}[S] = \mathbb{E}_{X_1}[\mathbb{E}_{S|X_1}[S|X_1]] = \mathbb{E}_{X_1}[X_1 + 3.5] = \mathbb{E}[X_1] + 3.5 = 3.5 + 3.5 = 7.$$

A.5 DATA: I.I.D. SAMPLES AND EXCHANGEABILITY

In statistical learning, we work with data sets that are modeled as collections of random variables. The standard assumption is that the data are independent draws from one fixed (but unknown) distribution.

Definition A.47 (i.i.d. random variables)**.** *Random variables $Z_1, \dots, Z_N$ with values in $\mathcal{Z}$ are called **independent and identically distributed (i.i.d.)** if they are independent and all have the same distribution, i.e., $\mathbb{P}_{Z_1} = \dots = \mathbb{P}_{Z_N}$. If this common distribution is $\mathbb{P}_Z$, we write $Z_1, \dots, Z_N \overset{i.i.d.}{\sim} \mathbb{P}_Z$.*

Remark A.48. *In supervised learning, $Z_n = (X_n, Y_n) \in \mathcal{X} \times \mathcal{Y}$, and the data set $(X_n, Y_n)_{n=1}^N$ is typically assumed to be i.i.d. with distribution $\mathbb{P}_{X,Y}$. A test point $(X_{test}, Y_{test}) \sim \mathbb{P}_{X,Y}$ is assumed to be independent of the data. Note that the X_n here denote different samples, and not the components of a vector-valued random variable as in some of the examples above.*

Conformal prediction relies on a weaker assumption than i.i.d., which only requires that the order of the data does not matter.

Definition A.49 (Exchangeability)**.** *Random variables $Z_1, \dots, Z_N$ with values in $\mathcal{Z}$ are called **exchangeable** if for every permutation σ of $\{1, \dots, N\}$, the random vector $(Z_{\sigma(1)}, \dots, Z_{\sigma(N)})$ has the same (joint) distribution as $(Z_1, \dots, Z_N)$.*

Proposition A.50. *If $Z_1, \dots, Z_N$ are i.i.d., then they are exchangeable.*

Proof. Let $\mathbb{P}_Z$ be the common distribution and σ a permutation. By independence, for all measurable $B_1, \dots, B_N \subseteq \mathcal{Z}$,

$$\mathbb{P}(Z_{\sigma(1)} \in B_1, \dots, Z_{\sigma(N)} \in B_N) = \prod_{n=1}^{N} \mathbb{P}(Z_{\sigma(n)} \in B_n) = \prod_{n=1}^{N} \mathbb{P}_Z(B_n) = \mathbb{P}(Z_1 \in B_1, \dots, Z_N \in B_N).$$

Hence the joint distributions of $(Z_{\sigma(1)}, \dots, Z_{\sigma(N)})$ and $(Z_1, \dots, Z_N)$ coincide on all product sets $B_1 \times \dots \times B_N$. One can show (using measure theory) that this implies that the two distributions coincide. If densities exist, this can also be seen directly: The joint density $\prod_{n=1}^N p_Z(z_n)$ is invariant under permutations of $(z_1, \dots, z_N)$. □

Example A.51 (Exchangeable, but not independent)**.** *An urn contains one red and one blue ball. We draw both balls one after the other without replacement, and denote by $Z_1, Z_2 \in \{\text{red}, \text{blue}\}$ the colors of the first and second ball. Then (Z_1, Z_2) is either $(\text{red}, \text{blue})$ or $(\text{blue}, \text{red})$, each with probability $1/2$. Swapping the two entries does not change this distribution, hence Z_1, Z_2 are exchangeable. However, they are not independent, since*

$$\mathbb{P}(Z_1 = \text{red}, Z_2 = \text{red}) = 0 \neq \frac{1}{4} = \mathbb{P}(Z_1 = \text{red})\mathbb{P}(Z_2 = \text{red}).$$

Remark A.52 (Outlook: why exchangeability matters)**.** *Let $Z_1, \dots, Z_{N+1}$ be exchangeable real-valued random variables that are almost surely pairwise distinct. By symmetry, each of the $N+1$ variables is equally likely to be the smallest, the second smallest, and so on. Hence, the rank of Z_{N+1} among $Z_1, \dots, Z_{N+1}$ is uniformly distributed on $\{1, \dots, N+1\}$. Consequently, for $k \in \{1, \dots, N\}$, the probability that Z_{N+1} does not exceed the k-th smallest of $Z_1, \dots, Z_N$ is $k/(N+1)$. Applied with $Z_1, \dots, Z_N$ the scores of the calibration data, Z_{N+1} the score of a test point, and $k = \lceil (N+1)(1-\alpha) \rceil$,*

this is the key argument behind the coverage guarantee of conformal prediction, which we will discuss in detail later.

BIBLIOGRAPHY

[1] Francis Bach. *Learning Theory from First Principles*. MIT Press, 2024.